10

Theory of line formation

10.1 Formation of optically thin lines

We have seen earlier that we always see a layer with $\tau_\lambda \approx \frac{2}{3}$, i.e., it is always $F_\lambda \approx B_\lambda$ ($\tau_\lambda = \frac{2}{3}$). Spectral lines occur at wavelengths for which κ_λ is very large. Therefore, we see much higher layers at these wavelengths (Fig. 10.1). These have a lower temperature and so B_λ is smaller. This leads to a smaller F_λ in the line than F_c, the continuum flux in the neighborhood of the line. We define the line depth R_λ (see Fig. 10.2) as

$$R_\lambda = \frac{F_c - F_\lambda}{F_c} = 1 - \frac{F_\lambda}{F_c}. \qquad 10.1$$

Clearly, R_λ depends on the wavelength within the line. The total area in the line divided by F_c (see the hatched area in Fig. 10.3) is called the *equivalent width* W_λ. The division by F_c means we are measuring the flux in units of the continuum flux, which means the continuum flux becomes 1. We then have the situation shown in Fig. 10.3: the equivalent width describes the width a rectangular spectral line must have in order to have the same total absorption as the actual line. From this definition we find

$$W_\lambda = \int_{\text{line}} R_\lambda \, \mathrm{d}\lambda = \int_{\text{line}} \frac{F_c - F_\lambda}{F_c} \, \mathrm{d}\lambda, \qquad 10.2$$

where the integral has to be extended over the whole line. In the following,

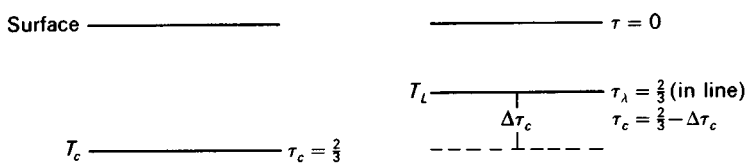

Fig. 10.1. In the wavelength of the line, where $\kappa_\lambda \gg \kappa_c$, we see much higher layers with $T = T_L$ for which $T_L < T_c$ and therefore $F_\lambda = B_\lambda(T_L) < F_c = B_\lambda(T_c)$, where T_c is the temperature of the layer seen in the continuum.

τ_c refers to the optical depth in the continuum, while τ_λ is the optical depth in the line; this means $\tau_\lambda = \tau_{\text{line}} + \tau_c$.

For weak lines, i.e., if $\kappa_{\text{line}} \ll \kappa_{\text{continuum}}$, the layer $\tau_\lambda = \frac{2}{3}$ will be close to the layer with $\tau_c = \frac{2}{3}$; we can make a Taylor expansion of $B_\lambda(\mathrm{T})$ around the layer for which $\tau_c = \frac{2}{3}$ and find

$$B_\lambda(\tau_\lambda = \tfrac{2}{3}) = B_\lambda(\tau_c = \tfrac{2}{3}) - \left(\frac{\mathrm{d}B_\lambda}{\mathrm{d}\tau_c}\right)_{\tau_c=2/3} \Delta\tau_c = B_\lambda(\tau_c = \tfrac{2}{3} - \Delta\tau_c). \qquad 10.3$$

It is

$$\frac{\tau_\lambda}{\tau_c} = \frac{\kappa_\lambda}{\kappa_c} \qquad \text{or} \qquad \tau_c = \tau_\lambda \frac{\kappa_c}{\kappa_c + \kappa_L}, \qquad 10.4$$

where $\kappa_L = \kappa_{\text{line}}$, the line absorption coefficient, and κ_c is the continuous absorption coefficient. It is $\kappa_\lambda = \kappa_c + \kappa_L$. For the layer with $\tau_\lambda = \frac{2}{3}$ we find

$$\tau_c = \frac{2}{3}\left(\frac{\kappa_c}{\kappa_c + \kappa_L}\right) = \frac{2}{3} - \Delta\tau_c. \qquad 10.5$$

For $\kappa_L/\kappa_c \ll 1$ the ratio

$$\frac{\kappa_c}{\kappa_c + \kappa_L} \approx 1 - \frac{\kappa_L}{\kappa_c} \qquad \text{and} \qquad \tau_c \approx \frac{2}{3}\left(1 - \frac{\kappa_L}{\kappa_c}\right) = \frac{2}{3} - \Delta\tau_c$$

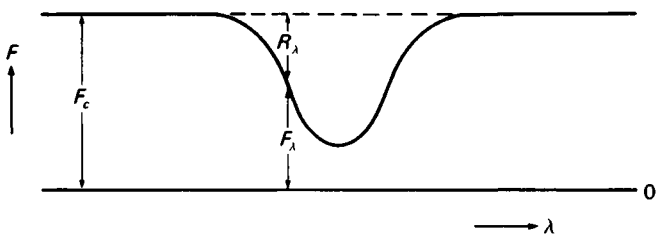

Fig. 10.2. A schematic plot of the energy distribution around a spectral line as a function of wavelength λ. At the wavelength of the line the flux F_λ is smaller than the flux in the continuum F_c. R_λ is the line depth measured in units of the continuum flux.

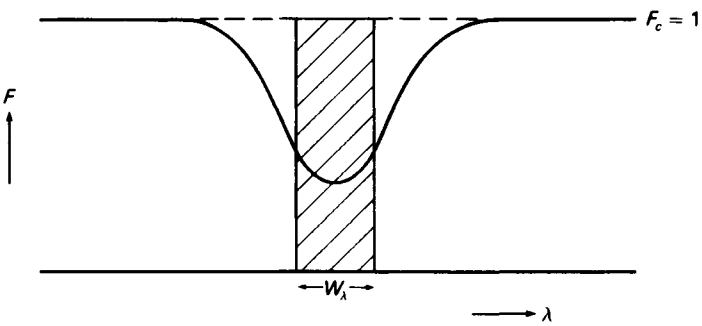

Fig. 10.3. A schematic plot of the energy distribution in the neighborhood of the line. The equivalent width W_λ gives the width which a rectangular spectral line must have in order to have the same total absorption as the actual line.

Making use of this, we see that

$$\Delta\tau_c = +\frac{2}{3}\frac{\kappa_L}{\kappa_c} \qquad \text{if } \kappa_L \ll \kappa_c. \tag{10.6}$$

If $\kappa_L \ll \kappa_c$, the line is called *optically thin*.

With equation (10.6) we find for the line depth

$$R_\lambda = \frac{F_c - F_\lambda}{F_c} = \frac{B_\lambda(\tau_c = \frac{2}{3}) - B_\lambda(\tau_\lambda = \frac{2}{3})}{B_\lambda(\tau_c = \frac{2}{3})}$$

$$= \frac{B_\lambda(\tau_c = \frac{2}{3}) - B_\lambda(\tau_c = \frac{2}{3}) + \left.\frac{\mathrm{d}B_\lambda}{\mathrm{d}\tau_c}\right|_{\tau_c = 2/3} \frac{2}{3}\frac{\kappa_L}{\kappa_c}}{B_\lambda(\tau_c = \frac{2}{3})}, \tag{10.7}$$

or

$$\boldsymbol{R_\lambda = \frac{2}{3}\frac{\kappa_L}{\kappa_c}\left.\frac{\mathrm{d}\ln B_\lambda}{\mathrm{d}\tau_c}\right|_{\tau_c = 2/3}}. \tag{10.8}$$

For an absorption tube we found (Section 4.4) $I_\lambda = I_{\lambda_0}\mathrm{e}^{-\tau_\lambda} \sim I_{\lambda_0}(1-\tau_\lambda)$, if $\tau_\lambda \ll 1$, and the line depth becomes

$$R_\lambda = \frac{I_{\lambda 0} - I_\lambda}{I_{\lambda 0}} = 1 - (1 - \tau_\lambda) = \tau_\lambda = \kappa_{L,\mathrm{cm}} H_{\mathrm{eff}}, \tag{10.9}$$

where H_{eff} is the length of the absorption tube.

Comparing equation (10.8) with (10.9), we can write (10.8) in the form

$$R_\lambda = \kappa_{L,\mathrm{cm}} H_{\mathrm{eff}} \qquad \text{with } H_{\mathrm{eff}} = \frac{2}{3}\left(\frac{\mathrm{d}\ln B_\lambda}{\mathrm{d}\tau_c}\right)_{\tau_c = 2/3} \frac{1}{\kappa_{c,\mathrm{cm}}}. \tag{10.10}$$

H_{eff} can be looked at as the effective height of the atmosphere with respect to line absorption, or in other words, the stellar atmosphere acts as a laboratory absorption tube of length H_{eff}.

Equation (10.8) is a very important relation. It tells us that the line depth is not only determined by the size of the line absorption coefficient, but also by the value for the continuous absorption coefficient and, last but not least, by the gradient of the source function, which can usually be replaced by the gradient of the Planck function. *If there is no temperature gradient with the temperature decreasing outwards, then there are no absorption lines in the spectrum.*

If we want to study the line profile, i.e., the energy distribution within the line, then we have to study the wavelength dependence of κ_L, as is obvious from equation (10.8).

10.2 The line absorption coefficient

The line absorption coefficient κ_L per cm^3 is given by the absorption coefficient per atom $\kappa_{L,a}$ (dimension cm^2) times the number of atoms per cm^3 absorbing in the line under consideration, N_L. In the case of the Balmer lines, for instance, N_L is given by the number of hydrogen atoms in the second quantum level N_{H} ($n = 2$). The line absorption coefficient $\kappa_{L,a}$ is not infinitely sharp as one might think, considering that the line is due to a transition of an electron from one bound state to another bound state, for which the frequency of the absorbed or emitted light is given by

$$h\nu = \chi \text{ (upper state)} - \chi \text{ (lower state)},$$

where χ is the excitation energy. $\kappa_{L,a}$ extends over a certain wavelength band. The reason for this natural width of the line is the natural width of the energy levels. In the classical approach of the oscillating electron, the emission in the line is the emission of a bound oscillating electron with eigenfrequency ν. Due to this radiation, the oscillation is damped and, as we saw before, the emitted energy has a frequency dependence given by the emissivity $\varepsilon_{L,a}$ (since $\varepsilon_{L,a} = \kappa_{L,a} S_\nu = \kappa_{L,a} B_\nu$ in LTE, the profile of $\kappa_{L,a}$ resembles that of $\varepsilon_{L,a}$). With ω_0 the frequency in the center of the line, one finds

$$\varepsilon_{L,a} \propto \frac{\gamma}{(\omega^2 - \omega_0^2)^2 + \gamma^2\omega^2} \sim \frac{\gamma}{(2\omega)^2(\omega - \omega_0)^2 + \gamma^2\omega^2} = \frac{\gamma}{(\Delta\omega^2 + \gamma^2/4)} \frac{1}{2^2\omega^2}. \tag{10.11}$$

Here we have used

$$\omega^2 - \omega_0^2 = (\omega - \omega_0)(\omega + \omega_0) \sim 2\omega(\omega - \omega_0) \qquad \text{for } \frac{\omega - \omega_0}{\omega} \ll 1.$$

The frequency $\omega = 2\pi\nu$ can be considered to be constant over the line. The important quantity for the line profile is the denominator. We find

$$\kappa_{L,a} \propto \frac{\gamma}{\Delta\omega^2 + (\gamma/2)^2}. \tag{10.12}$$

This so-called *damping profile*, or *Lorentzian profile*, has a total half width $2\Delta\omega(\frac{1}{2}) = \gamma$ (see Fig. 10.4). The larger the amount of radiation, the larger and the broader the line.

From the classical damping theory of the oscillating and radiating electron one finds

$$\gamma = \frac{8\pi^2 e^2}{3mc\lambda_0^2}, \tag{10.13}$$

where λ_0 is the center wavelength of the line.

The natural total half width of the line for a classical electron comes out to be $\lambda_N = 1.2 \times 10^{-4}$ Å, independently of the wavelength λ.

For the absorption coefficient $\kappa_{L,a}$ we find

$$\kappa_{L,a} = \frac{\pi e^2}{mc} \frac{\gamma}{\Delta\omega^2 + (\gamma/2)^2} f. \qquad 10.14$$

Here we have introduced the factor f, which is called the *oscillator strength* because this factor indicates how many classical oscillators the line under consideration corresponds to.

In the quantum-mechanical approach, the natural width of the line is determined by the finite width of the two energy levels. According to the Heisenberg uncertainty principle, the uncertainty of an energy measurement ΔE is given by the time Δt which is available for the measurement, namely

$$\Delta E \, \Delta t = \hbar = h/2\pi, \qquad h = \text{Planck's constant}. \qquad 10.15$$

The maximum time available for the measurement of the energy E is given by the time which the electron stays in that energy level. The larger the transition probability, the shorter the lifetime, the stronger the radiation and the larger ΔE, which means the larger the natural line width. It turns out then that γ is proportional to the transition probability. This means that the natural line width is of the order

$$\Delta\omega/\omega = \Delta\lambda/\lambda \propto \text{transition probability}. \qquad 10.16$$

Actually, it is the sum over all the transition probabilities for the level under consideration. When calculating the line width we obviously have to take into account the widths of both levels which are involved in the line transition.

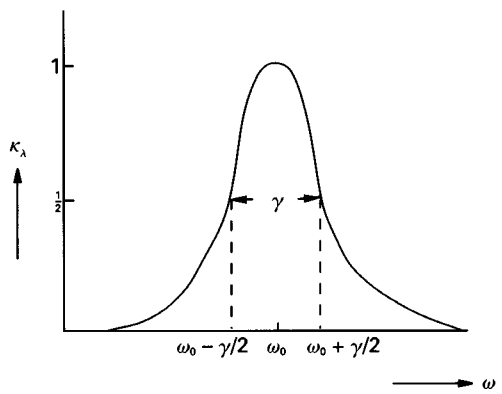

Fig. 10.4. A schematic plot of the classical damping profile for an oscillating electron with eigenfrequency ω_0 and a damping constant γ.

In most astrophysical cases the line width is, however, determined not by the radiation damping, but by the thermal motion of the emitting and absorbing atoms.

10.3 The Doppler profile

The light emitting atoms in the stellar atmosphere are not at rest, but have a Maxwell velocity distribution which for one velocity component $\xi = v_x$ is given by

$$\frac{dN}{N} = \frac{1}{\sqrt{\pi}} e^{-(\xi/\xi_0)^2} \frac{d\xi}{\xi_0} \qquad \text{where } \zeta_0^2 = \frac{2R_g T}{\mu}. \qquad 10.17$$

A particle moving with ξ away from the observer (see Fig. 10.5) emits light at a wavelength $\omega_0 + \Delta\omega$ with

$$\frac{\Delta\omega}{\omega_0} = -\frac{\xi}{c} \qquad \text{or} \qquad \frac{\Delta\omega}{\omega_0} = \frac{\xi}{c} \qquad 10.18$$

for motion away from the observer for motion towards the observer

Only the component of the motion in the direction of the line of sight has to be considered for the Doppler effect. Therefore, we have to consider the Maxwell distribution for one velocity component only.

The number of emission or absorption processes at $\omega + \xi\omega/c$ is then given by the number of atoms moving with a velocity component ξ towards us. This means that the emission or absorption profile must be given by

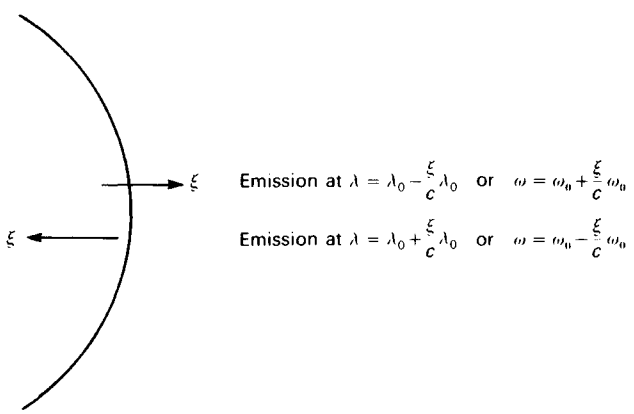

Fig. 10.5. Thermal motions of amount ξ of the light emitting particles leads to emission at frequencies $\omega = \omega_0 \pm \omega_0\xi/c$ seen by us, the observers. $\omega = \omega_0$ in the rest frame of the particle.

the Maxwell distribution (10.17), or

$$\kappa_{L,a} \propto N(\xi) \propto \mathrm{e}^{-(\xi/\xi_0)^2} \frac{1}{\xi_0} \propto \mathrm{e}^{-(\Delta\omega/\Delta\omega_D)^2} \frac{1}{\Delta\omega_D}, \qquad 10.19$$

with

$$\frac{\xi}{c} = \frac{\Delta\lambda}{\lambda} = -\frac{\Delta\omega}{\omega} \quad \text{and} \quad \frac{\xi_0}{c} = \frac{\Delta\lambda_D}{\lambda} = -\frac{\Delta\omega_D}{\omega} \quad \text{and} \quad \xi_0 = \sqrt{\left(\frac{2R_gT}{\mu}\right)},$$

or

$$\kappa_{L,a} \propto \mathrm{e}^{-(\Delta\lambda/\Delta\lambda_D)^2} \frac{1}{\Delta\lambda_D} \quad \text{with } \Delta\lambda_D = \lambda \frac{\xi_0}{c}. \qquad 10.20$$

For an electron which would actually radiate like a classical oscillator, one obtains

$$\int_{-\infty}^{\infty} \kappa_\nu \mathrm{d}\Delta\nu = \frac{\pi e^2}{mc} \quad \text{or} \quad \int_{-\infty}^{\infty} \kappa_\lambda \mathrm{d}\Delta\lambda = \frac{\lambda^2}{c}\frac{\pi e^2}{mc}, \quad \text{since } \mathrm{d}\lambda = -\frac{c}{\nu^2}\mathrm{d}\nu. \qquad 10.21$$

With the classical radius of the electron being given by

$$r_{\mathrm{el}} = \frac{e^2}{mc^2},$$

we can also write

$$\int_{-\infty}^{\infty} \kappa_\lambda \mathrm{d}\Delta\lambda = \lambda^2 \pi r_{\mathrm{el}}.$$

The integral over κ_λ over the line must, of course, remain the same, independently of the Doppler broadening. In a frequency independent radiation field the amount of energy absorbed by the atoms does not change when the atoms are in motion. The condition (10.21) then leads to the value for $\kappa_{L,a}$:

$$\kappa_{L,a} = \frac{\sqrt{\pi} e^2}{mc^2} \frac{\lambda^2}{\Delta\lambda_D} \mathrm{e}^{-(\Delta\lambda/\Delta\lambda_D)^2}. \qquad 10.22$$

Actually, the radiation for electrons in different energy states and for different transitions is different in intensity, which is not obvious from equation (10.21). As mentioned earlier, one therefore introduces a factor f describing the actual strength of the lines. This factor tells us to how many classical oscillators the emission corresponds. Therefore, we call this factor the oscillator strength of the line, or the transition.

For broadening by the Doppler effect only, we find a half width

$$\Delta\lambda_{1/2} = \Delta\lambda_D \sqrt{(\ln 2)}. \qquad 10.23$$

For the sun with $T \sim 6000$ K we find for an iron atom with atomic weight $\mu = 56$ that

$$\xi_0 = \sqrt{\left(\frac{2RT}{\mu}\right)} = \sqrt{\left(\frac{2 \times 8.3 \times 10^7 \times 6 \times 10^3}{56}\right)} \approx 1.4 \times 10^5 \text{ cm s}^{-1},$$

or

$$\xi_0 = 1.4 \text{ km s}^{-1} \quad \text{and} \quad \frac{\Delta\lambda_D}{\lambda} = \frac{1.4 \times 10^5}{3 \times 10^{10}} \sim 4 \times 10^{-6}.$$

For $\lambda \sim 4000$ Å the $\Delta\lambda_D \sim 1.6 \times 10^{-10} \sim 0.016$ Å, i.e., the Doppler width is much larger than the damping width of the line, which is about 1.1×10^{-4} Å, as we saw above.

For the hydrogen atom at $T = 6000$ K, we find

$$\xi_0 = \sqrt{\left(\frac{2 \times 8.3 \times 10^7 \times 6 \times 10^3}{1}\right)} \approx 10 \text{ km s}^{-1}.$$

This is always a good reference point to estimate average thermal velocities. The Doppler width of the Balmer lines is about 0.1 Å. However, in most of the stars the hydrogen lines are much wider because of Stark effect broadening, which we shall discuss in Chapter 11.

10.4 The Voigt profile

Generally, we have to consider both effects: the Doppler profile and the damping profile. The damping profile is negligible in the line center because the Doppler width is so much larger, but in the wings the Doppler profile decreases very steeply, while the damping profile decreases only as $1/\Delta\lambda^2$. The general profile, also called the '*Voigt*' *profile*, consists therefore of a Doppler core and damping wings (see Fig. 10.6). The final form of the profile depends on the ratio of the damping widths $\gamma/2$ to the Doppler width $\Delta\omega_D$. This ratio is usually denoted by α, i.e., $\alpha = \gamma/2\Delta\omega_D$. The line absorption coefficient κ_L is then given by

$$\kappa_{L,a} = \kappa_0 H(\alpha, v) \quad \text{with} \quad \kappa_0 = \frac{\sqrt{\pi} \cdot e^2}{mc^2} \frac{\lambda^2}{\Delta\lambda_D} \cdot f. \qquad 10.24$$

In Table 10.1 we tabulate the function from which the Voigt profiles can be calculated.

In Fig. 10.7 we show a logarithmic plot of line profiles for different values of α. As a rule of thumb we can say that we start to see the damping wings at a distance $\Delta\lambda$ from the line center for which

$$\Delta\lambda = -(\log \alpha)\Delta\lambda_D. \qquad 10.25$$

This rule is often useful for a first guess of the Doppler widths $\Delta\lambda_D$.

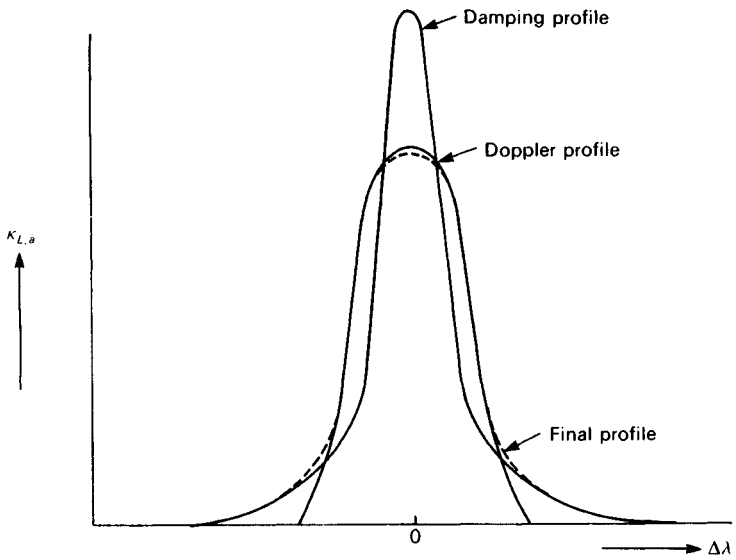

Fig. 10.6. A schematic plot of the different line profiles contributing to the Voigt profile. Generally, the line profile consists of the Doppler core and the damping wings. It is called a Voigt profile.

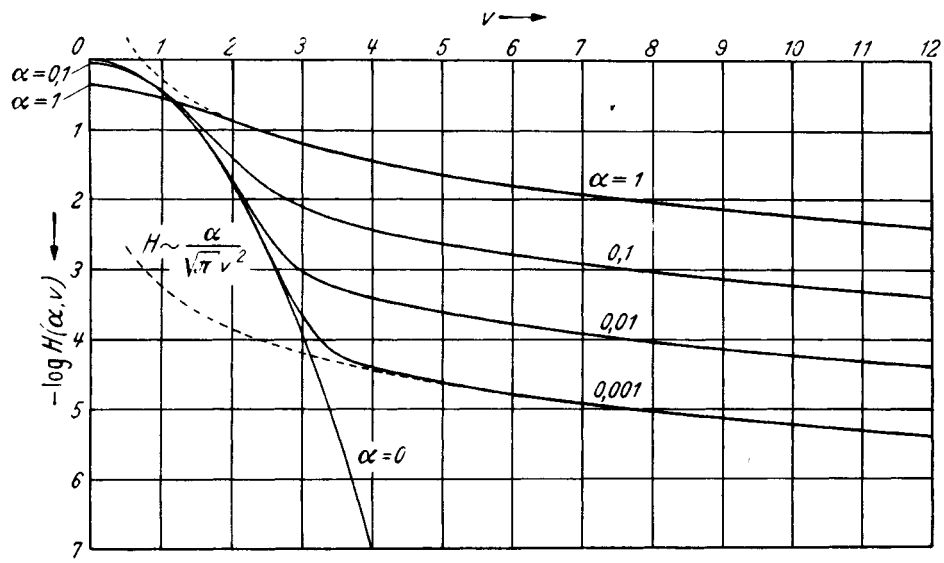

Fig. 10.7. Voigt profiles for different values of α are shown in a logarithmic plot. (From Unsöld, 1955, p. 265.)

Table 10.1. *Functions needed to calculate the Voigt function $H(\alpha, v)$ for $\alpha \ll 1$ according to D. L. Harris (1948)*

$$v = \frac{\Delta\lambda}{\Delta\lambda_D} = \frac{\Delta\omega}{\Delta\omega_D} \qquad \alpha = \frac{\gamma}{2\Delta\omega_D}$$

$$H(\alpha, v) = e^{-v^2} - \frac{2\alpha}{\sqrt{\pi}}[1 - 2vF(v)] \quad \text{with} \quad F(v) = e^{-v^2}\int_0^v e^{u^2}\,du$$

v	e^{-v^2}	$-\frac{2}{\sqrt{\pi}}[1 - 2vF(v)]$
0.0	+1.000 000	−1.128 38
0.1	+0.990 050	−1.105 96
0.2	+0.960 789	−1.040 48
0.3	+0.913 931	−0.937 03
0.4	+0.852 144	−0.803 46
0.5	+0.778 801	−0.649 45
0.6	+0.697 676	−0.485 52
0.7	+0.612 626	−0.321 92
0.8	+0.527 292	−0.167 72
0.9	+0.444 858	−0.030 12
1.0	+0.367 879	+0.085 94
1.1	+0.298 197	+0.177 89
1.2	+0.236 928	+0.245 37
1.3	+0.184 520	+0.289 81
1.4	+0.140 858	+0.313 94
1.5	+0.105 399	+0.321 30
1.6	+0.077 305	+0.315 73
1.7	+0.055 576	+0.300 94
1.8	+0.039 164	+0.280 27
1.9	+0.027 052	+0.256 48
2.0	+0.018 3156	+0.231 726
2.1	+0.012 1552	+0.207 528
2.2	+0.007 9071	+0.184 882
2.3	+0.005 0418	+0.164 341
2.4	+0.003 1511	+0.146 128
2.5	+0.001 9305	+0.130 236
2.6	+0.001 1592	+0.116 515
2.7	+0.000 6823	+0.104 739
2.8	+0.000 3937	+0.094 653
2.9	+0.000 2226	+0.086 005
3.0	+0.000 1234	+0.078 565
3.1	+0.000 0671	+0.072 129
3.2	+0.000 0357	+0.066 526
3.3	+0.000 0186	+0.061 615
3.4	+0.000 0095	+0.057 281

Table 10.1 *continued*

v	e^{-v^2}	$-\frac{2}{\sqrt{\pi}}[1-2vF(v)]$
3.5	+0.000 0048	+0.053 430
3.6	+0.000 0024	+0.049 988
3.7	+0.000 0011	+0.046 894
3.8	+0.000 0005	+0.044 098
3.9	+0.000 0002	+0.041 561
4.0	+0.000 0001	+0.039 250
4.2	—	+0.035 195
4.4	—	+0.031 762
4.6	—	+0.028 824
4.8	—	+0.026 288
5.0	—	+0.024 081

10.5 Line broadening due to turbulent motions

The absorption or emission of radiation by the atoms is, of course, also Doppler shifted if other motions are present in addition to the thermal motions in the atmosphere. We saw, for instance, the Doppler shifts due to the solar granulation motion, as seen in the wiggly lines of the solar spectrum (see Volume 1, Chapter 18). If the sun were very far away and we could only take a spectrum of the light integrated over the whole sun the lines would no longer be wiggly, but would be correspondingly broadened. We would call this *macroturbulent broadening*, meaning that *different parts of the stellar disc have different velocities*. If we assume that these macroturbulent velocities have a velocity distribution similar to the distribution of the thermal velocities, i.e., Gaussian, then the resulting velocity distribution obtained after folding the macroturbulent velocity distribution with the thermal velocity distribution is again a Gaussian distribution with a reference velocity ξ_0, which is given by

$$\xi_0^2 = \xi_{\text{th}}^2 + \xi_{\text{turb}}^2, \qquad 10.26$$

and the Doppler width of the line then corresponds to $\Delta\lambda_D$, which is now determined by the ξ_0 given in equation (10.26). ξ_{th} stands for the reference velocity of thermal motions and ξ_{turb} is the reference velocity of the turbulent motions.

In addition to the macroturbulent motions, we may also have so-called *microturbulence*. If along the line of sight down to $\tau_\lambda = \frac{2}{3}$ we encounter different velocities, then the photons coming out of the atmosphere

encounter atoms with different velocities, just as in the case of the thermal motions. In this case it appears to the photons as if the absorption coefficient $\kappa_{L,a}$ is broadened. It appears as if the thermal motions are enhanced. The lines are broadened just as in the case of macroturbulence, but the effect on the total line absorption in a given line is different than in the case of macroturbulence. In the case of macroturbulence, think of the solar granulation, each part of the stellar disc is independent of the others. For instance, a line in a granulum becomes optically thick for a given number of absorbing atoms, no matter whether the matter in the granulum is moving up or down or whether it is at rest. Within each granulum the line profile and the total absorption does not change. It can easily be seen that when you add up all the spectra for the many granules the total absorption does not change either. Only the apparent line profile becomes wider because the lines from the different granules appear at different wavelengths. This is different for microturbulence. Since along the line of sight the atoms now have different velocities, a photon with a small wavelength shift with respect to the center of the line which would have been reabsorbed on the way out will now possibly not be reabsorbed again because the material on top may be moving out very fast, and the absorption coefficient at the wavelength of the photon is now very small. At any given wavelength within the line the absorption coefficient appears to be altered. The κ_L/κ_c at each wavelength within the line is changed. For an increasing number of absorbing atoms the line remains optically thin much longer than if no microturbulence were present. This has important consequences when we look at the total absorption in such lines.

10.6 Other distortions of the line profiles

Another intrinsic distortion of the line profiles in stars is due to stellar rotation, as we discussed in Volume 1, Chapter 13.

As demonstrated in Fig. 10.8, one half of the star moves away from us, while the other half moves towards us. The component of motion in the direction of the line of sight is $v_y = v_r(\vartheta) \sin \phi$, if $v_r(\vartheta)$ is the rotational velocity for the colatitude ϑ. For rigid body rotation $v_r(\vartheta) = v_e \sin \vartheta$, and the velocity component v_y along the line of sight is $v_y = v_e \sin \phi \sin \vartheta$, where v_e is the equatorial rotational velocity. v_y is constant for all points on the sphere for which $\sin \phi \sin \vartheta = \text{const}$. Where these points are on the sphere can best be seen if we describe the sphere of the star in spherical polar

coordinates. It is (see Fig. 10.9):

$$x = \rho \sin \phi = r \sin \vartheta \sin \phi$$
$$y = \rho \cos \phi = r \sin \vartheta \cos \phi$$
$$z = r \cos \vartheta .$$

This shows that $\sin \vartheta \sin \phi$ is constant for $x = \text{const.}$, i.e., for stripes across the star (see Fig. 10.10). A spectral line broadened by rotation only has an elliptical line profile (see Fig. 10.11) if all areas on the stellar surface are equally bright.

The spectral line profile is generally also distorted by the finite resolution of the spectrograph, which gives a line profile a certain shape, even if the spectral line were intrinsically infinitely sharp. This line profile is called the

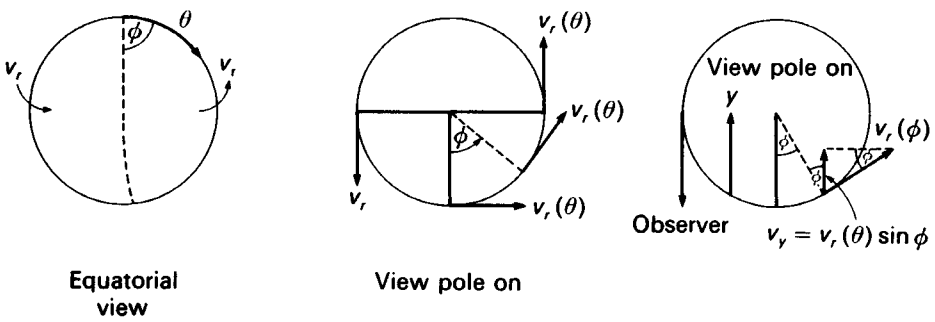

Fig. 10.8. One half of a rotating star moves towards us. The other half moves away from us.

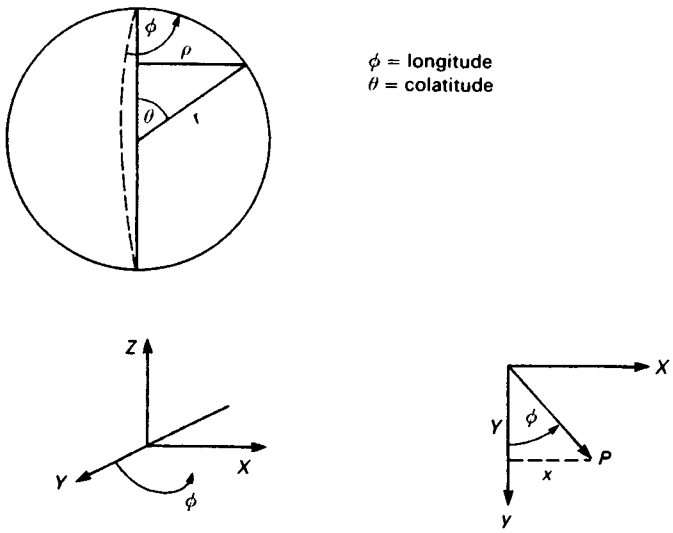

Fig. 10.9 shows the relation between the cartesian coordinates X, Y, Z and the spherical polar coordinates, r, ϑ, ϕ.

instrumental profile. It gives an additional line broadening, especially for low dispersion spectra. For such low dispersion spectra the instrumental broadening is frequently more important than any broadenings intrinsic to the star. On such low dispersion spectra we cannot therefore measure intrinsic line profiles. For high dispersion spectra we have to correct the measured line profile for the instrumental broadening, which has to be measured in the laboratory. Quite often we are unable to extract the true line profiles, especially for spectra of faint stars, for which we can only get low resolution spectra. We can then only work with the total absorption in the lines. The total absorption in the lines remains unchanged due to all these macroscopic broadening mechanisms. (This does not prevent us

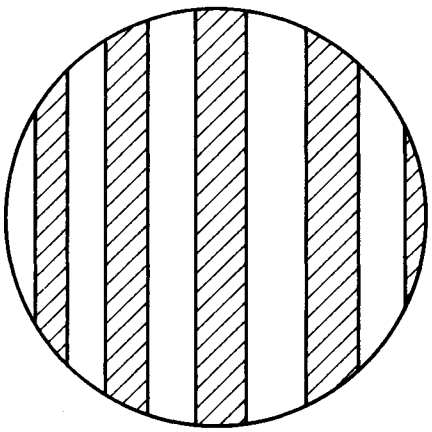

Fig. 10.10. v_y = const. for stripes on the star's disc parallel to the projected axis of rotation.

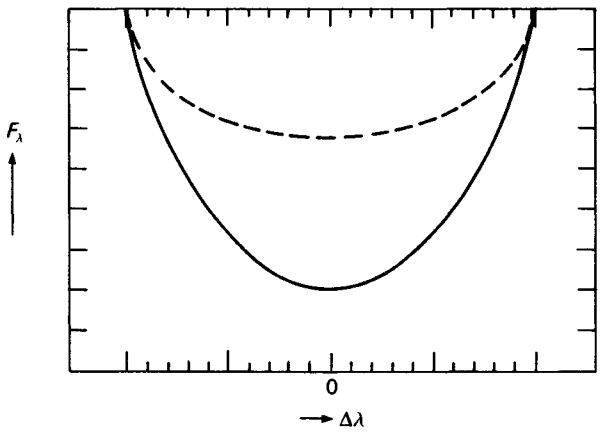

Fig. 10.11. A spectral line broadened by rotation only has an elliptical line profile for a star with no limb darkening (---). (——) profile with strong limb darkening. (From Gray, 1976, p. 398.)

from measuring slightly wrong values for lower dispersion spectra because we tend to make wrong judgements for the position of the continuum on such low dispersion spectra.)

10.7 Equivalent widths for optically thin lines

The equivalent width for an optically thin line can now easily be calculated, making use of equations (10.2) and (10.8). We find

$$\begin{aligned} W_\lambda &= \int_{\text{line}} R_\lambda \, \mathrm{d}\lambda = \frac{2}{3}\frac{1}{\kappa_c}\frac{\mathrm{d}\ln B_\lambda}{\mathrm{d}\tau_c}\int_{\text{line}} \kappa_{\text{line}} \, \mathrm{d}\lambda \\ &= \frac{2}{3}\frac{1}{\kappa_c}\frac{\mathrm{d}\ln B_\lambda}{\mathrm{d}\tau_c} N_{rs} \int_{\text{line}} \kappa_{\text{line}} \text{ (per atom)} \, \mathrm{d}\lambda \\ &= C N_{rs} \lambda^2 \frac{\pi e^2}{mc^2} f_{rs} = C' \lambda^2 N_{rs} f_{rs}. \end{aligned} \qquad 10.27$$

Here r is the index indicating the degree of ionization of the absorbing ion or atom and the second index s describes the quantum numbers of the two levels involved in the transition. Equation (10.27) tells us that for optically thin lines the equivalent width is proportional to the number of absorbing atoms times the oscillator strength. With

$$N_{rs} = N_{r0} \frac{g_{rs}}{g_{r0}} \mathrm{e}^{-\chi_s/kT},$$

we find that

$$W_\lambda \propto \frac{N_{r0}}{g_{r0}} g_{rs} f_{rs} \lambda^2 \mathrm{e}^{-\chi_s/kT}. \qquad 10.28$$

If we measure W_λ and $g_{rs} f_{rs}$ in the laboratory as well as χ_s, we can determine N_{r0}/g_{r0} and thereby the number of atoms or ions of the particular element for which the line was observed. However, in order to make use of the Boltzmann formula we must know the temperature. We have already discussed several methods for the determination of atmospheric temperatures. In this context it seems to be best to determine the temperature directly from the study of lines which originate from different levels with different excitation energies. If we measure the W_λ for two such lines, one with index 1 and another with index 2, we find for the equivalent widths

$$\begin{aligned} W_{\lambda_1} &\propto \frac{N_{r0}}{g_{r0}} g_{r_1} f_1 \lambda_1^2 \mathrm{e}^{-\chi_1/kT} \\ W_{\lambda_2} &\propto \frac{N_{r0}}{g_{r0}} g_{r_2} f_2 \lambda_2^2 \mathrm{e}^{-\chi_2/kT}. \end{aligned} \qquad 10.29$$

The ratio of the equivalent widths gives

$$\frac{W_{\lambda_1}}{W_{\lambda_2}} = \frac{g_{r_1} f_1}{g_{r_2} f_2} \frac{\lambda_1^2}{\lambda_2^2} e^{-(\chi_1 - \chi_2)/kT}. \qquad 10.30$$

If the excitation energies are known, the temperature can be determined. This so determined temperature is called the *excitation temperature* because it is determined from the excitation difference of the two levels. We can also say that we determine T in such a way that both lines give the same N_{r_0}/g_{r_0}, as must be the case.

All these derivations only hold as long as $\kappa_{\text{line}}/\kappa_c \ll 1$. For most spectral lines observed this is not true. In fact, all lines with $W_\lambda \geqslant 15$ mÅ are not strictly optically thin lines.

10.8 Optically thick lines

Equation (10.8) tells us the shape of the line profile in the optically thin case. How does the line look when the line becomes optically thick? Fig. 10.12 demonstrates schematically the line profiles for increasing values of κ_L/κ_c or with increasing number of absorbing atoms. As long as $\kappa_L/\kappa_c \ll 1$, i.e., for optically thin lines, the line depth at every point within the lines increases proportional to κ_L/κ_c, or proportional to the number of atoms N_s absorbing in the line. This is no longer the case when $\kappa_L/\kappa_c \approx 1$ or greater than 1. In this case, the lines are called optically thick. When lines become optically thick they necessarily reach a maximum depth R_λ, because R_λ cannot become larger than 1 (we cannot have less than zero intensity). Actually, the lowest intensity in the line center is given by $S_\lambda(\tau_\lambda = \frac{2}{3})$ in the line center. For optically very thick lines for which κ_L/κ_c in the line center goes to infinity, the intensity in the line center is given by the source function $S_\lambda(\tau_c = 0)$. The lowest central intensity in the line is therefore given by $S_\lambda(0)$ or in LTE by $B_\lambda(0)$ which is not zero, since $T(0)$ is not zero (see equation

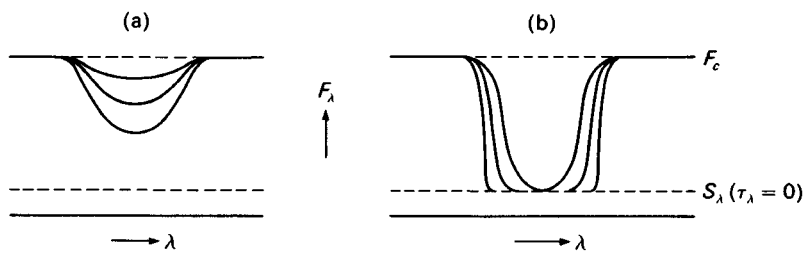

Fig. 10.12. Changes of the line profile with increasing κ_L/κ_c for (a) optically thin and (b) optically thick lines.

8.30). (In cases of NLTE when $S_\lambda \neq B_\lambda$ the $S_\lambda(0)$ may tend towards zero, for instance, in resonance lines, but these are exceptional cases, although usually the strongest lines.)

In the wings the line is still optically thin. There the line depth still grows proportionally to κ_L/κ_c, so the line becomes wider, and the profile becomes more rectangular. Since the line flanks are very steep, the increase in total absorption is very small with increasing values of κ_L/κ_c. The increase in optical thickness thus gives an additional line broadening which might be mistaken for an additional broadening due to turbulence, especially if the line is broadened by rotation or by the instrumental profile such that the line depth may not appear to be very deep (see, for instance, Volume 1, Fig. 13.3, where we show an optically very thick line in a non-rotating star and in a star with high rotational velocity). It appears similar to an optically thin line with very high turbulent velocities. It is not always easy to distinguish between these possibilities. The best way seems to be by Fourier analysis, although extremely accurate measurements are necessary.

10.9 The curve of growth

10.9.1 Interpolation of the line profile

In this section we will discuss how the total absorption within the spectral line, measured by the equivalent width W_λ (see Section 10.1), is related to the line absorption coefficient κ_L and thereby to the number of atoms absorbing in a given line. Once we understand this relation we will be able to determine the number of absorbing atoms, and from that determine the chemical composition of the stars.

The total absorption in the line is given by

$$W_\lambda = \int_{\text{line}} \frac{F_c - F_\lambda}{F_c}\, \mathrm{d}\lambda = \int_{\text{line}} R_\lambda\, \mathrm{d}\lambda, \qquad 10.31$$

where the integral has to be extended over the whole line. In the previous sections we have discussed the wavelength dependence of R_λ for optically thin lines. We have also seen that for optically thick lines the limiting central line depth is given by R_{λ_0} with

$$R_{\lambda_0} = \frac{F_c - S_\lambda(0)}{F_c}. \qquad 10.32$$

In order to describe the whole line profile, we need an interpolation for R_λ which is equal to equation (10.8) in the part of the line which is optically thin, i.e., in the wings of the line, and which equals equation (10.32) in the

optically thick parts of the line. Such an interpolation was given by Minnaert. We introduce a new quantity

$$X(\lambda) = \frac{2}{3}\frac{\kappa_L(\lambda)}{\kappa_c}\frac{\mathrm{d}\ln B_\lambda}{\mathrm{d}\tau_c}, \qquad 10.33$$

with the derivative taken at $\tau_c = \frac{2}{3}$, and with $\kappa_L(\lambda_0) = \frac{\sqrt{\pi}e^2}{mc^2}\frac{\lambda_0^2}{\Delta\lambda_D}N_s f$, and N_s = number of absorbing atoms per cm^3.

According to equation (10.8) we find for optically thin lines that $R_\lambda = X(\lambda)$. We also see that in the line center of optically thick lines where R_λ is given by equation (10.32), the R_{λ_0} is much smaller than $X(\lambda_0)$. The quantity $X(\lambda)$ can obviously become much larger than 1, while R_λ cannot. So for

$$X(\lambda) \gg 1 \qquad \text{we find} \qquad R_\lambda(\lambda) \ll X(\lambda).$$

With this in mind, we can now obtain a very good approximation for the line profile by the interpolation formula

$$R_\lambda = \frac{1}{\dfrac{1}{X(\lambda)} + \dfrac{1}{R_{\lambda_0}}}. \qquad 10.34$$

For very large values of κ_L/κ_c the $X(\lambda)$ is much larger than 1. It is then also much larger than R_{λ_0} such that $1/X(\lambda)$ is much smaller than $1/R_{\lambda_0}$ and we have $R_\lambda = R_{\lambda_0}$.

For small $X(\lambda)$, i.e., optically thin lines, $1/X(\lambda) \gg 1/R_{\lambda_0}$ and $R_\lambda \sim X(\lambda)$. As we saw before for a Doppler profile

$$X(\lambda) = X(\lambda_0 + \Delta\lambda) = X(\lambda_0)\mathrm{e}^{-(\Delta\lambda/\Delta\lambda_D)^2}, \qquad 10.35$$

with

$$X(\lambda_0) = \frac{\sqrt{\pi}e^2}{mc^2}\frac{\lambda_0^2}{\Delta\lambda_D}N_s f H_{\mathrm{eff}}. \qquad 10.36$$

The integral

$$W_\lambda = \int_{\text{line}} R_\lambda\,\mathrm{d}\lambda = \int_{\text{line}} \frac{1}{\dfrac{1}{X(\lambda)} + \dfrac{1}{R_{\lambda_0}}}\,\mathrm{d}\lambda$$

can be computed. What we find is a relation between W_λ and $X(\lambda_0)$, $\Delta\lambda_D$, and R_{λ_0}.

The relation between W_λ and $X(\lambda_0)$ is called the *curve of growth*. For a Doppler profile it depends on the two parameters R_{λ_0} and $\Delta\lambda_D$.

10.9.2 The curve of growth

The curve of growth describes how the equivalent width W_λ depends on $X(\lambda_0)$. Since $X(\lambda_0) \propto N_s$, the number of absorbing atoms, this is equivalent to saying how W_λ depends on the number of atoms absorbing in the lines for a given κ_c and a given $\mathrm{d} \ln B_\lambda / \mathrm{d}\tau_c$.

For all optically thin lines, we saw above that the line depth at any point within the line grows proportionally to κ_L/κ_c, or proportionally to N_s. The integral over the line depth, which is equal to W_λ, therefore also grows proportionally to N_s. For large $X(\lambda_0)$ the growth of W_λ becomes much smaller, since no absorption is added any more in the line center. The line only increases very slowly in width. The equivalent width therefore increases only very slowly. We find

$$W_\lambda \propto \sqrt{(\ln X(\lambda_0))} \propto \sqrt{(\ln N_s f)}. \qquad 10.37$$

This part of the curve is called the *flat part* of the curve of growth (see Fig. 10.13). If $X(\lambda_0)$ becomes very large, the line becomes so strong that even the weak, far out wings due to the damping become visible. Then the equivalent width increases again and one finds

$$W_\lambda \propto \sqrt{(N_s \gamma f)}, \qquad 10.38$$

where γ is the damping constant (see Section 10.2).

This part is called the *damping part*, or the *square root part*, of the curve of growth.

The extent of the flat part of the curve of growth depends very much on the ratio $\gamma/(2\Delta\lambda_D) = \alpha$, as seen in Fig. 10.13, which shows schematic curves of growth.

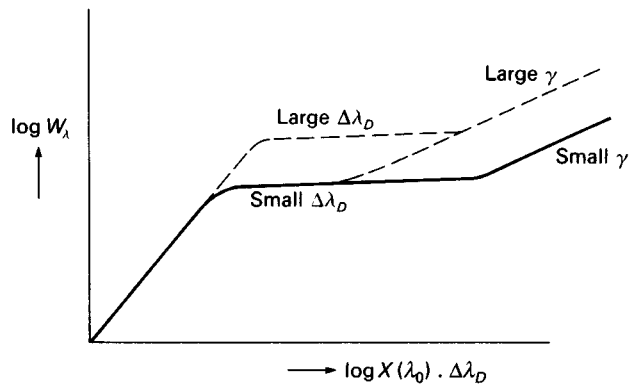

Fig. 10.13. Schematic curve of growth for different values of the Doppler width $\Delta\lambda_D$ and damping constants γ.

10.9.3 The influence of the Doppler width on the curve of growth

We still have to discuss how the Doppler width enters into the curve of growth.

Fig. 10.14 shows that for increasing line width saturation occurs for larger W_λ; in fact, for the same degree of saturation or the same depth in the line center the equivalent width increases proportionally to the line width, i.e., proportionally to $\Delta\lambda_D$: so the flat part of the curve of growth is higher by a factor proportional to $\Delta\lambda_D$. This is demonstrated in Figs. 10.13 and 10.14. However, if in the curve of growth we plot $\log(W_\lambda/\Delta\lambda_D)$, then the level of the flat part of the curve of growth remains independent of $\Delta\lambda_D$. In order to keep the linear part of the curve of growth in place, we have to divide the abscissa by $\Delta\lambda_D$ as well and plot $\log(W_\lambda/\Delta\lambda_D)$ as a function of $\log(X(\lambda_0)$. We then obtain one curve of growth for all values of $\Delta\lambda_D$ (except for the damping part), which is quite useful if we want to do spectrum analysis by means of the curve of growth.

10.9.4 The influence of $S_\lambda(0)$ on the curve of growth

From Fig. 10.15 it is obvious that for very strong lines the flat part of the curve of growth will increase proportionally to the central line depth R_{λ_0}, since the line profile is almost rectangular. Plotting $\log(W_\lambda/R_{\lambda_0}\Delta\lambda_D)$ as a function of $\log(X(\lambda_0)/R_{\lambda_0})$ will give us a curve of growth which is independent of R_{λ_0} and $\Delta\lambda_D$.

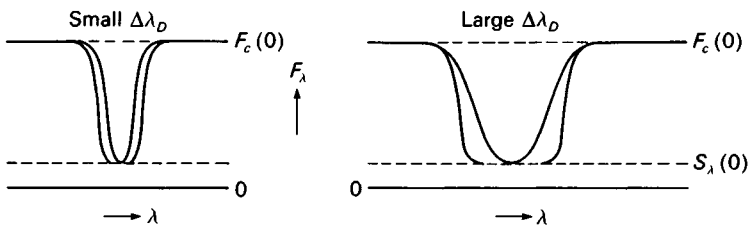

Fig. 10.14. For the same central line depth, the equivalent widths of optically thick lines increase proportionately to the line widths.

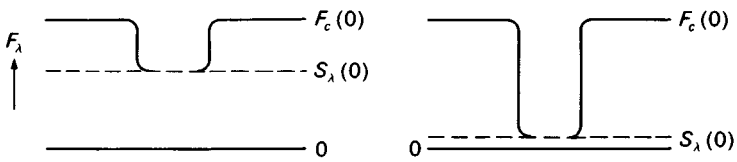

Fig. 10.15. For saturated lines the equivalent width becomes proportional to the central line depth.

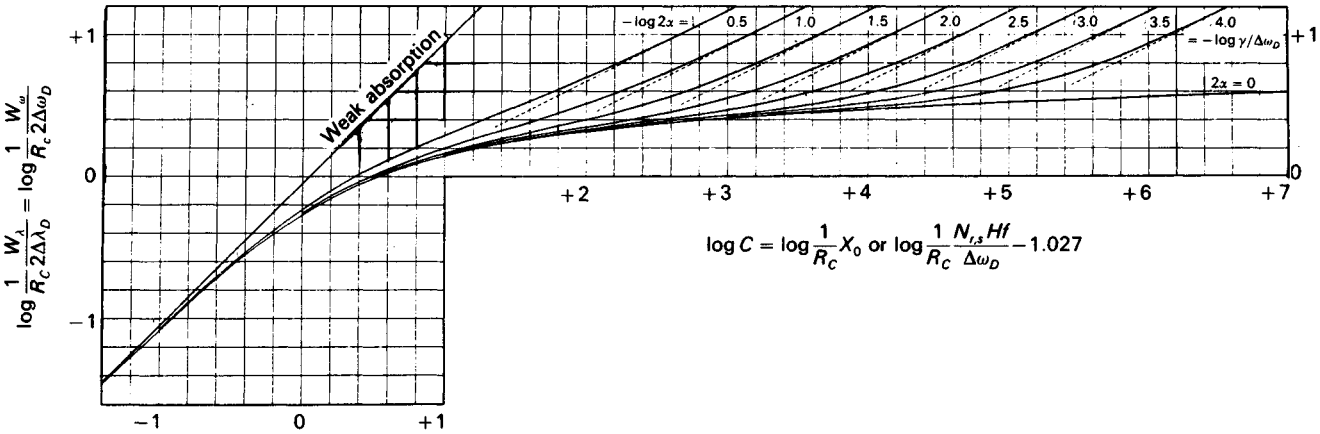

Fig. 10.16. The so-called Universal curves of growth according to Unsöld, 1955, depending only on $\alpha = (\gamma/2\Delta\omega_D)$. $R_c = R(\lambda_0)$ and $X_0 = X(\lambda_0)$.

This is the form of the curve of growth which Unsöld calls the 'Universal' curve of growth, which we show in Fig. 10.16.

10.9.5 The influence of the damping constant on the curve of growth

We still have to talk about the influence of the damping constant γ on the damping part of the curve of growth. Clearly, the damping wings will be more pronounced if γ is larger. Therefore, for larger γ the damping part of the curve of growth will start for smaller values of $X(\lambda_0)$. The value of γ is usually expressed by means of the parameter α, with $\alpha = \gamma/2\Delta\omega_D$. The larger α, the earlier the damping part will start. Fig. 10.16 shows the final curves of growth obtained for different values of α, as given by Unsöld (1955).

11

The hydrogen lines

The hydrogen lines deserve special consideration, not only because they are the strongest lines, but also because they are a very important tool in analyzing stellar spectra. The reason is that they are broadened by the so-called 'molecular' Stark effect, that is by the electric fields due to the passing ions. When a radiating atom or ion finds itself in an electric field, the energy levels are shifted to slightly different values of the excitation energy χ_n or χ_{exc}. The emitted lines therefore occur at slightly different wavelengths. Even more important is that for the hydrogen atom the different orbitals contributing to the energy level with a main quantum number n, are normally degenerate, i.e., when there is no external force field they all have the same excitation energy (this is, of course, indicated by the statistical weight for each level). In the presence of an external force field this degeneracy is removed, so that in an external field the different orbitals contributing to a given level with main quantum number n now occur at slightly different values for the excitation energy. This in turn means that, instead of one line being emitted in the force free case, there are now several lines emitted or absorbed at slightly different wavelengths. Fig. 11.1 shows the different components which are observed for the different Balmer lines.

This shifting of the energy levels in the electric field of the neighboring ions has mainly two effects. First, it *shortens the lifetime of the undisturbed level* which, according to equation (10.15), *results in a broadening of the undisturbed energy level.* Since this broadening is due to the electric field of the ions passing nearby or due to 'collisions' with other ions, this broadening is called *impact broadening.* Since this broadening is also due to a shortening of the lifetime of the original level, its effect on the width of the line is the same as that of radiation damping. It leads to a line profile which is similar to that for radiation damping but with a larger γ. Since

the collision with the passing ion only takes a short time, the new, shifted levels only have a very short lifetime and are also broadened by this effect.

In addition to this impact broadening of the unshifted lines, we have to consider the effects of the new line components, for which the splitting increases with increasing field strengths. Ions passing very close cause a large splitting, while ions passing at larger distances only cause a small splitting. The pattern of the absorbed or emitted line therefore depends on the statistics of the electric fields which the absorbing or emitting atoms experience at the moment of the absorption or emission of the photon. In order to calculate the overall profile of the absorbed line, we have to look at the statistics of the electric fields experienced by the atoms at the moment

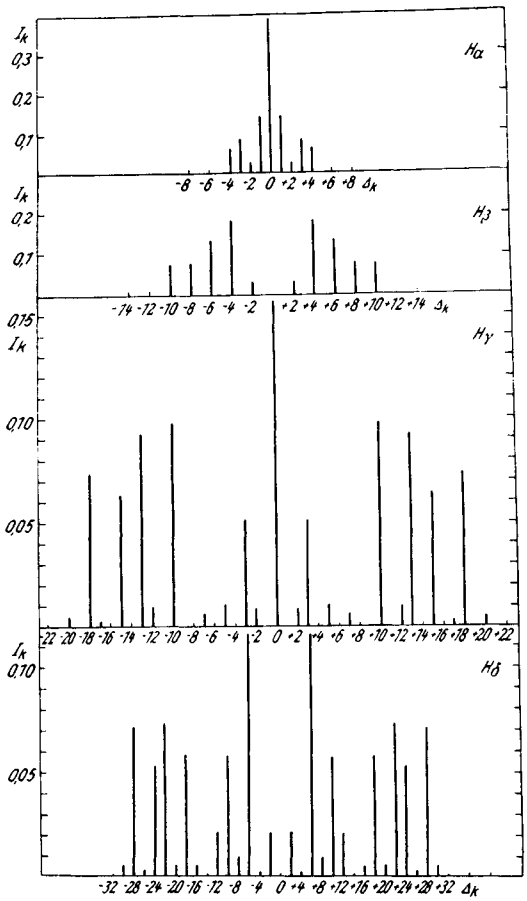

Fig. 11.1. The Stark effect splitting of the different Balmer lines (according to Unsöld, 1955, p. 320.)

of absorption. This part of the line broadening is described by the so-called *statistical line broadening theory*. Since it deals with the lines originating from the shifted energy levels, they occur in the wings of the original line. The larger the electrical fields or, in other words, the closer the encounters, the further out in the wing this displaced line occurs. The line profile in the wings of the hydrogen lines is therefore described by the statistical theory of the broadening by the 'molecular' Stark effect. It is clear that this broadening is not actually due to molecules, but rather to ions. This name is only used to indicate that it is not due to macroscopic electric fields, but rather to microscopic fields due to the charged particles in the plasma.

If we want to calculate the distribution of the absorption coefficient for the hydrogen lines, we have to calculate the probability for absorption at a given wavelength difference $\Delta\lambda$ from the line center. If this $\Delta\lambda$ is still within the impact broadened Lorentzian profile of the original line, then we know the absorption coefficient from the Lorentzian line profile; but if $\Delta\lambda$ is larger and is in the wings where the statistical broadening theory has to be applied, we have to determine the probability of finding an ion at the correct distance d from the absorbing atom that creates a field $F = e/d^2$ leading to the shift $\Delta\lambda$ under consideration. We see immediately that the chance of finding a particle at the required distance d is larger if there are more particles per cm^3. The chance of finding at a given moment an ion at exactly the required distance is proportional to the number of particles N^+ present per cm^3. We therefore find that in the case of the statistical line broadening in the hydrogen wings the absorption coefficient at any given $\Delta\lambda$ is proportional to the number of ions per cm^3 which equals the number of free electrons per cm^3, at least if all ions are ionized only once.

Why do we never mention the electric fields caused by the passing electrons? We said that the broadening is due to the shifting of energy levels in the atom due to the electric field experienced by the absorbing atom. In the case of the passing electrons the velocities of the electrons are too large. The collision times are too short. In other words, the phase shift $\eta = \int \Delta\omega \, dt$ experienced by a light wave is too small.

The actual profiles of the hydrogen line absorption coefficients have been calculated in successively better approximations. The latest ones were tabulated by Vidal, Cooper and Smith (1973). What is of interest to us here is the fact that at any $\Delta\lambda$ in the wings of the hydrogen lines κ_λ is proportional to the number of free electrons per cm^3, n_e. For high density atmospheres when n_e is large, the hydrogen lines are much broader than for low density atmospheres in supergiants (see Fig. 11.2).

Another important fact is the increase in the splitting of the different Stark effect components of the lines with increasing main quantum number n (see Fig. 11.1). In fact, the splitting increases with n^2. For a given electric field the Balmer lines with a larger n of the upper level are broadened more than Hα or Hβ. On the other hand, we know that as the upper energy levels of the hydrogen atom get closer and closer together, the energy difference decreases as $1/n^3$. For some level $n = n_u$ the line broadening becomes larger than the difference in wavelength for successive Balmer lines. The lines merge together and can no longer be seen separately (see Fig. 11.2). From a simplified broadening theory for the Balmer lines – the original Holtsmark theory – Inglis and Teller (1939) find that the highest quantum number n_u for which the energy levels are still separated is related to the electron density n_e by

$$\log n_e = 23.26 - 7.5 \log n_u. \qquad 11.1$$

This equation provides a very simple means of determining the electron density n_e. We only have to count the number of visible Balmer lines, determine the upper quantum number of the last visible line n_u, and calculate n_e from equation (11.1).

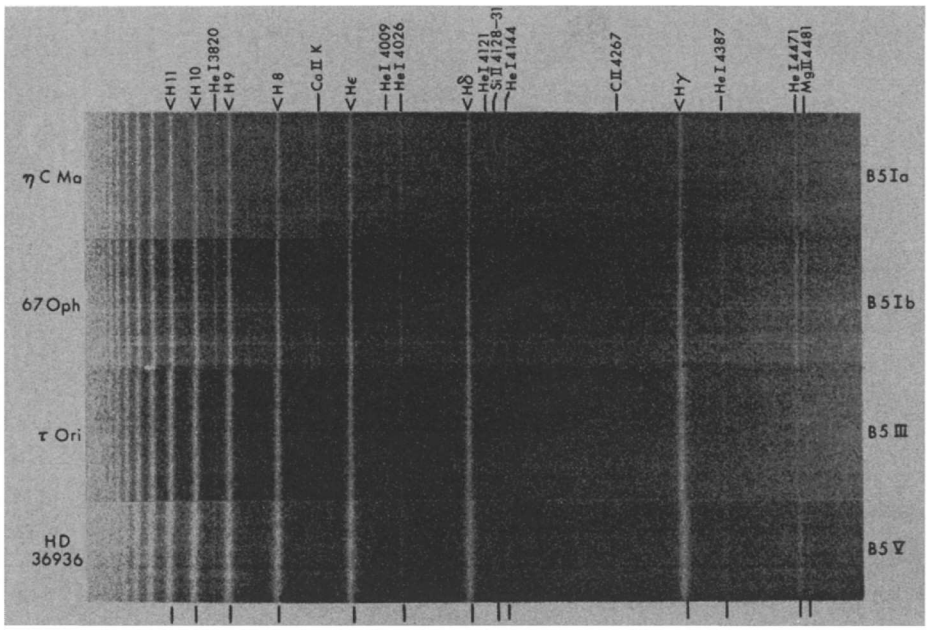

Fig. 11.2. The spectra of a B5 main sequence star and a B5 supergiant are compared. In the main sequence star the hydrogen lines are much wider than in the supergiant. For the supergiant we can see more Balmer lines separately. (From Morgan, Abt and Tapscott, 1978. This is a negative. White means no light.)

12

Spectrum analysis

12.1 The Balmer jump and the hydrogen lines

12.1.1 Hot stars

In Section 8.2 we saw that the Balmer discontinuity is determined by the ratio of the continuous absorption coefficients on both sides of this discontinuity. For hot stars we derived that the discontinuity is determined by the ratio of the number of hydrogen atoms in the second quantum level (absorbing shortward of 3647 Å) to the number of hydrogen atoms in the third quantum level (absorbing on the long wavelength side of 3647 Å). This ratio is completely determined by the temperature. For the B stars the temperature can in principle be determined from the Balmer discontinuity alone.

The pressure can be determined from the electron density n_e. For late B stars, for instance, the hydrogen is completely ionized, while the helium is not. Therefore, we know that $n_e = \mathrm{H}^+$ and $N = n_e + \mathrm{H}^+ + \mathrm{He} = n_e + \mathrm{H}^+ + 0{\cdot}1\mathrm{H}^+$ if the abundance of helium is 10% by number of atoms. With this we find

$$N = 2.1 n_e, \qquad 12.1$$

and, of course,

$$P_g = NkT.$$

As we saw in the previous chapter, the number of free electrons can be determined from the hydrogen lines in two ways: either from the number of visible Balmer lines by means of the Inglis–Teller formula (equation 11.1), or by means of the hydrogen line wings. In Section 10.1 we saw that the line depth for optically thin lines is given by equation (10.8):

$$R_\lambda = \frac{2}{3}\frac{\kappa_L}{\kappa_c}\frac{\mathrm{d}\ln B_\lambda}{\mathrm{d}\tau_c}. \qquad 12.2$$

For the hot stars the continuous absorption coefficient κ_c in the visual is due to the Paschen continuum, which means that it is due to absorption

from the third level of the hydrogen atom. This means κ_c is proportional to $N_H(n=3)$. The line absorption coefficient for the Balmer lines which originate by absorption from the second quantum level is, of course, proportional to the number of hydrogen atoms in the second quantum level $N_H(n=2)$. As we saw in the previous chapter, the line absorption coefficient in the wings of the Balmer lines is also proportional to the number of free electrons because of the Stark effect broadening of these lines. Therefore, we find for the hydrogen lines in the B stars

$$R_\lambda \propto \frac{N_H(n=2)n_e}{N_H(n=3)}. \qquad 12.3$$

Knowing the temperature from the Balmer discontinuity, the depth in the Balmer line wings tells us the electron pressure and thereby the gas pressure.

Since the gas pressure must balance the weight of the overlying material, the gas pressure also gives a rough estimate for the gravitational acceleration in the stars. We must have approximately

$$P_g = mg, \qquad 12.4$$

where m is the mass of the overlying material.

Knowing that we are looking down into an optical depth $\tau_c = \frac{2}{3}$ and also knowing that the continuous absorption is due to the hydrogen atoms in the third quantum level, we can calculate the number of hydrogen atoms in the third quantum level above $\tau_c = \frac{2}{3}$ from

$$hN_H(n=3) = \frac{2}{3}\frac{1}{\kappa_{ca}}, \qquad 12.5$$

where κ_{ca} is the continuous absorption coefficient per hydrogen atom in the level $n=3$, and h is the effective height of the overlying atmosphere.

From the Boltzmann and Saha formula we can calculate the number of H^+ ions which constitute the main fraction of the mass. Considering the abundance of helium the total mass per H^+ ion is given by $m_H(1+0.1\times 4) = 1.4m_H$ and the total mass m above the level $\tau_c = \frac{2}{3}$ is given by

$$m = h \times 1.4 \times N_{H^+} \times m_H. \qquad 12.6$$

An approximate value for the gravity g is then obtained from

$$g = P_g/m. \qquad 12.7$$

In determining the mass per H^+ ion we have already made use of the abundance of the elements which we only know after we have made a complete analysis. We will discuss this shortly.

In Section 8.2 we discussed the dependence of the Balmer discontinuity on pressure and temperature. We saw that for B stars the size of the

discontinuity is determined by

$$\frac{\kappa(3647^+)}{\kappa(3647^-)} = \frac{\kappa^+(\mathrm{H})N_{\mathrm{H}}(n=3)}{\kappa^-(\mathrm{H})N_{\mathrm{H}}(n=2)} \propto \frac{N_{\mathrm{H}}(n=3)}{N_{\mathrm{H}}(n=2)} = \frac{g_3}{g_2}\,\mathrm{e}^{-(\chi_3-\chi_2)/kT},$$

which means the discontinuity depends *only* on the temperature, which can therefore be determined from the Balmer jump alone. However, for O stars and for supergiants the importance of the electron scattering introduces some dependence on the electron density.

12.1.2 The cool stars

In the cool stars the continuous absorption coefficient is determined by the negative hydrogen ion H^-. We discussed in Section 8.2 that for these stars the Balmer discontinuity is dependent on both the temperature and the electron pressure, because the absorption on the long wavelength side of the Balmer jump is due to H^- and the number of H^- ions increases with increasing electron density as given by the Saha equation.

Let us now look at the depth of the Balmer line wings in these stars. We again start from equation (12.2). The continuous absorption is now due to H^-, and the line absorption coefficient in the Balmer line wings is again proportional to the number of hydrogen atoms in the second quantum level, and to n_e, because of the Stark effect broadening. We thus have

$$R_\lambda \propto \frac{N_{\mathrm{H}}(n=2)n_e}{N(\mathrm{H}^-)} \propto \frac{N_{\mathrm{H}}(n=2)n_e}{N_{\mathrm{H}}(n=1)n_e\varphi(T)} = \frac{N_{\mathrm{H}}(n=2)}{N_{\mathrm{H}}(n=1)\varphi(T)}. \qquad 12.8$$

Here we have made use of the Saha equation, which tells us that the number of H^- ions is proportional to $N_{\mathrm{H}}(n=1)n_e$ times a function φ of T. For cool stars the depth in the Balmer line wings is only a function of temperature. We have for cool stars the inverse situation from what is found in hot stars. Now the Balmer discontinuity depends on T and n_e and the Balmer line wings depend on temperature alone. This is shown by the spectra in Fig. 12.1.

Of course, these considerations apply only to normal stars. For very luminous stars the Balmer lines sometimes show emission cores. For such stars we see immediately that our simple 'normal' atmosphere theory does not apply. Also, we have not talked about O stars, for which electron scattering becomes very important and for which our assumption that the source function equals the Planck function needs checking. It appears that this assumption is not true for these hot stars. A basic discussion of NLTE effects, including the derivation of the source function in general, will be given in Chapter 14.

For normal stars we have seen in this section that the Balmer jump and the hydrogen line wings together always determine the electron density and the temperature. We do not need to know anything else. What we determine are, of course, some average values for the atmosphere. In Fig. 12.2 we summarize the situation.

12.2 The Strömgren colors

In order to make use of these fortunate circumstances, Strömgren (1963) developed a color system which is especially designed to measure the Hβ line wings and the Balmer jump. In order to determine the

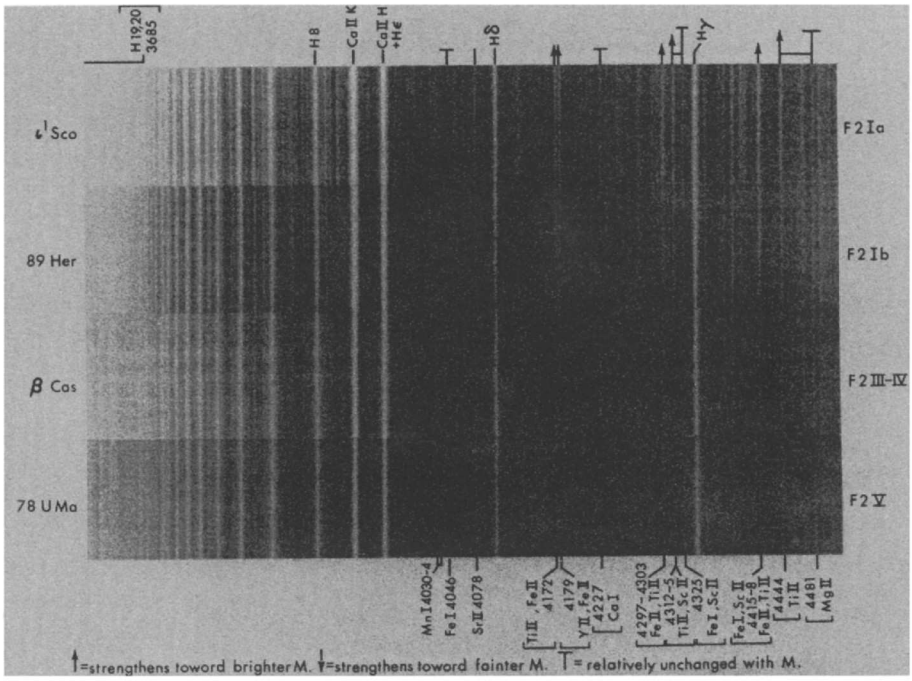

Fig. 12.1. For F stars the widths of the hydrogen lines are independent of the luminosity of the stars. (From Morgan, Abt and Tapscott, 1978.)

	Balmer discontinuity D	Hydrogen line wings H_β, H_λ, *etc.*
Hot stars	$D = D(T)$	$R_\lambda = R_\lambda(n_e, T)$
Cool stars	$D = D(n_e, T)$	$R_\lambda = R_\lambda(T)$

Fig. 12.2 summarizes the dependences of the Balmer jump and the depths of the hydrogen line wings on temperature and electron density for hot and for cool stars.

Table 12.1. *Central wavelengths and bandwidths of intermediate bandwidth photometry used by Strömgren*

Band	λ (Å)	Halfwidth (Å)	Band	λ (Å)	Halfwidth (Å)
u	3500	300	b	4670	180
v	4110	190	y	5470	230

Table 12.2. *Central wavelengths and bandwidths used in the Strömgren narrow-band photometry*

	Filter	λ_{max} (Å)	Halfwidth (Å)
a	Interference filter	5030	90
b	Interference filter	4861	35
c	Interference filter	4700	100
d	Interference filter	4520	90
e	Interference filter	4030	90
f	Glass filter	3650	350

abundances of the heavy elements, he also included a blue band which is particularly sensitive to the metallic line absorption. The wavelength bands used in the Strömgren photometry are much narrower than UBV color wavelength bands. With Strömgren photometry not quite such faint stars can be reached as with the UBV photometry, but better physical information can be obtained. Strömgren actually uses two color systems: one with intermediate bandwidths ($\Delta\lambda \approx 200$ Å) and one with still narrower bands ($\Delta\lambda \approx 100$ Å). The names, central wavelengths and bandwidths for both systems are given in Tables 12.1 and 12.2.

The Balmer jump (see Fig. 12.3) is found in both systems by measuring the intensity gradient on the long wavelength side, which would be about the expected gradient across the Balmer jump if there were no discontinuity, and comparing that with the actually measured gradient across the Balmer jump. In the intermediate band system the Balmer jump index is defined as

$$c_1 = \underbrace{(\mathrm{u} - \mathrm{v})}_{\text{actual gradient across the Balmer discontinuity}} - \underbrace{(\mathrm{v} - \mathrm{b})}_{\text{long wavelength gradient}}$$

where the letters stand for the magnitudes measured with the corresponding filters.

In the narrow band system the Balmer jump index is defined as

$$c = (\mathrm{f} - \mathrm{e}) \qquad - (\mathrm{e} - \mathrm{d})$$

gradient across the Balmer discontinuity — long wavelength gradient

The strength of the Balmer lines, which is determined by the strengths of the wings, is measured only in the narrow band system. Strömgren defines the l index as

$$l = b - \tfrac{1}{2}(\mathrm{a} + \mathrm{c}),$$

which means he measures the flux in the Hβ band with Hβ ($\lambda = 4861$ Å) in the center and compares it with the average intensity on both sides (see Fig. 12.4).

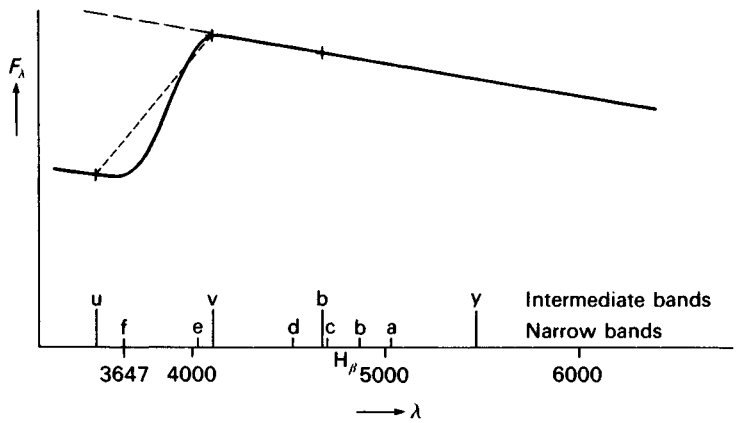

Fig. 12.3. The Strömgren bands describe the gradient of the energy distribution on both sides of the Balmer jump. The comparison of the extrapolation from the long wavelength side with the actual gradient over the discontinuity measures the Balmer jump.

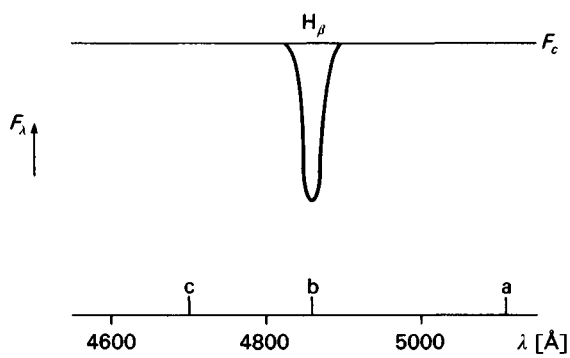

Fig. 12.4. The Hβ index compares the average flux and both sides of Hβ with the flux in Hβ.

Crawford (1958) modified this method and measured the flux through two filters, both centered at Hβ, but one with a bandwidth of 15 Å containing only intensities within the broad Hβ line, and one filter with a bandwidth of 150 Å, in which most of the light comes from wavelengths on both sides of Hβ. He defines an Hβ index as

$$\beta = (m(15\ \text{Å}) - m(150\ \text{Å})).$$

In the literature one usually finds the use of β and c_1, indicating a combination of both systems.

Our previous discussion of the Balmer lines tells us that these two indices determine T and n_e for all 'normal' stars.

The Strömgren color system is able to measure minor differences in the Balmer discontinuity. For cool stars rather minor differences in the electron density can therefore be measured, which is especially useful for the determination of stellar ages, as we shall discuss in Volume 3.

In the intermediate bandwidth system the b − y colors can serve to replace the B − V colors and can also be used to determine the temperature of the stars.

The overall strengths of the metallic lines, which can be used as a measure of the overall metal abundances, can be measured in the intermediate bandwidths system by the m_1 index defined as

$$m_1 = (\mathrm{v} - \mathrm{b}) - (\mathrm{b} - \mathrm{y}).$$

This index makes use of the fact that the line intensities are stronger in the blue and ultraviolet spectral region than in the yellow and red. It compares the intensity gradient at the shorter wavelengths, where it is reduced by the absorption lines, with the intensity gradient at longer wavelengths, where it is influenced very little by the lines.

In the narrow band photometry a similar index m can be defined by

$$m = (\mathrm{e} - \mathrm{d}) - (\mathrm{d} - \mathrm{a}),$$

which makes use of the same facts.

There are some problems with a simple interpretation of the m and m_1 indices. As we saw in the previous chapter, the line strength on the flat part of the curve of growth may be increased due to microturbulence. It is not quite clear how much the m index might be influenced by this effect. Stars with large microturbulence or with magnetic fields could possibly be mistaken for stars with a very large abundance of heavy elements. All color systems designed to measure heavy element abundances from the integral effects of many strong lines suffer from these limitations.

12.3 The curve of growth analysis

12.3.1 Line identification

In order to determine abundances, we have to identify the lines, i.e., we have to know to which elements they belong and to which transition they correspond. Therefore, first we have to measure the wavelengths of the lines under consideration. This is done by measuring the positions of lines of 'laboratory' spectra, also called 'comparison' spectra, which are photographed together with the stellar spectrum. These laboratory spectra are actually generated in a light source right at the telescope, such that the spectra can be observed through the same spectrograph and photographed and developed on the same photographic plate as the stellar spectrum (see Fig. 12.5). The wavelengths of the spectral lines in the laboratory spectra are predetermined. The positions of the stellar spectral lines then have to be measured accurately. The wavelengths of the stellar lines can then be determined by interpolation between the known wavelengths of the laboratory spectral lines.

Once we know the wavelengths, we have to identify the lines. There are tables that give all the wavelengths of the known lines and also give the corresponding transitions. Such a table is Miss Moore's *Multiplet Tables* (1959). At the back of these tables, we find a list of lines ordered according to wavelength. Once we have the measured wavelength of the line under consideration to within a certain uncertainty, which is determined by the dispersion and the general quality of the spectrum (the uncertainty is usually of the order of 0.1 Å if you have a high resolution spectrum with about 4 Å per mm), there are usually about three to four lines in that given uncertainty interval of about 0.1 Å between which we have to choose. For

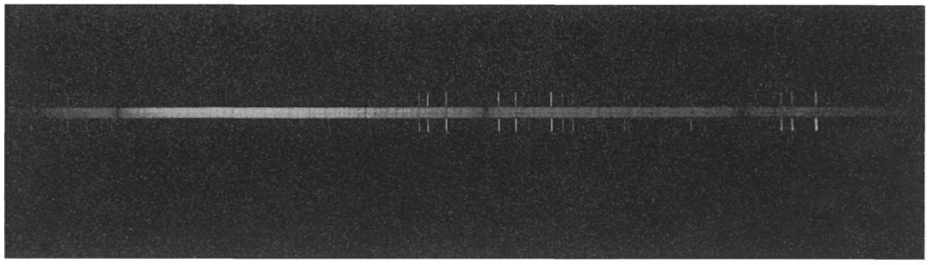

Fig. 12.5. Photograph of the stellar spectrum and of 'laboratory' spectra of an iron arc on both sides. The wavelengths for the spectral lines in the 'laboratory' spectrum have been determined previously. The wavelengths of the stellar lines can be determined by accurate interpolation.

each line in the table the line identification is given according to the element and to transition. This means the number of the multiplet is given to which this line belongs. In the multiplet tables one can then look up what the excitation energies are for the two levels involved in the transition and the statistical weights for the levels can also be obtained.

What is a multiplet?

In the hydrogen atom with only one electron all the orbitals with the same main quantum number n have the same energy. The energy levels are degenerate. In more complicated atoms with many electrons this is not the case. For the same electron configuration we can find different values for the total angular momentum J, depending on how the angular momenta of the different electrons are oriented with respect to each other. For a given orbital angular momentum L and a given spin S we still obtain different energy levels, depending on the value of the total angular momentum J. Each energy level with given values of L and S still splits up into several fine structure levels according to the orientation of L and S. Now consider two levels with different values of L and S. For each level we have the fine structure splitting according to the J values. All the lines originating from transitions between these fine structure levels of the two main levels are called one multiplet (see Fig. 12.6). For the lines in one multiplet the excitation energies of the different fine structure levels are

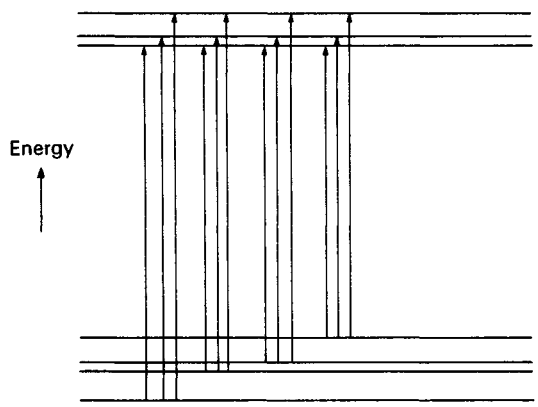

Fig. 12.6. For given values of the total orbital momentum L and the spin S the atomic energy level splits up into several fine structure levels according to the different values of J which depends on the relative orientation of S with respect to L. Lines due to transitions between the fine structure levels of two main levels are called a multiplet. All the transitions belonging to one multiplet are shown. Many of these transitions are 'forbidden', however, which means the transition probabilities are extremely small.

nearly equal. The relative line strengths for the lines within each multiplet are therefore completely determined by the transition probabilities or by their relative oscillator strengths. The relative intensities are therefore independent of temperature and pressure. This fact is very important for the spectrum analysis by means of the curve of growth.

Let us go back to the problem of line identification. Suppose the measured wavelength of a line at 3943.28 Å, which you want to identify, almost agrees with the wavelength of an iron line which belongs to the multiplet with the number 72. In order to check the identification you have to go to the front of the multiplet tables and look up the multiplet number 72 of iron. You find there are three lines in this multiplet which are stronger than the line you have measured. If your identification of the line under consideration with the iron line is correct, you should also see these three stronger lines in your spectrum at λ3949.9, 3977.74 and 4003.71 Å. They must be stronger than the line you have measured. You have to find the wavelengths of those three lines in your spectrum and see whether these lines are there and whether they are indeed stronger than the line you have measured. If they are, then the identification is possibly correct. If the three lines are not there or are too weak, then the identification of the measured line as this iron line is most probably wrong and you should check whether the line next to it in the multiplet table – namely, the Sm II line at 3943.239 – might be a better possibility. Only if all the relative line strengths within the multiplets in question match the expectations can you be relatively certain that you have made the correct identification.

12.3.2 Fitting the lines of one multiplet on the curve of growth

In principle the curve of growth analysis consists of measuring the equivalent width, dividing by $\Delta\lambda_D$ and by R_c, the maximum central depth of the strongest lines, and determining the ordinate in the curve of growth for a given line. We can then fit this value on the theoretical curve of growth shown in Fig. 10.16. The point where it fits on the curve of growth determines the abscissa. Because the abscissa depends on the number of absorbing atoms the value obtained for the abscissa tells us what the number of absorbing atoms is. The only problem in this procedure is that we do not know *a priori* what the Doppler width $\Delta\lambda_D$ is. Depending on the assumption about $\Delta\lambda_D$, we may fit a point for a strong line on a very different part of the curve of growth, as shown in Fig. 12.7. We therefore have to determine the value for $\Delta\lambda_D$ very carefully. Only for very weak lines on the linear part of the curve of growth do we find the same number

of absorbing atoms for a given equivalent width independently of the assumed value for the Doppler widths, because abscissa and ordinate are divided by the same factor, but such weak lines are hard to measure.

In addition, we would also like to determine the degree of excitation in the atoms and ions, in order to determine the total number of particles of a given kind and not just the number of particles absorbing in a given line.

In the following discussion we want to show how the Doppler width can also be determined by means of the curve of growth and to show how the degree of excitation can be determined. We start by looking at the lines of one multiplet only.

Once you know the correct line identification you have to look up the oscillator strengths of the lines in another table. Oscillator strengths can either be calculated by means of quantum mechanics which in the case of complicated atoms is difficult and not very accurate. For the hydrogen atom theoretical oscillator strengths are very well determined. For the complicated atoms like iron measurements appear to be more accurate, though the laboratory light sources usually show temperature variations within the light source, which makes it difficult to determine the degree of excitation which determines the number of emitting atoms. If this number was determined inaccurately then the measured oscillator strengths are all wrong. Because of this problem the first measured iron oscillator strengths for iron lines with higher excitation potentials were all off by nearly a factor of ten and correspondingly the measured iron abundances were all wrong by the same factor. We only say this to show that the measurement of

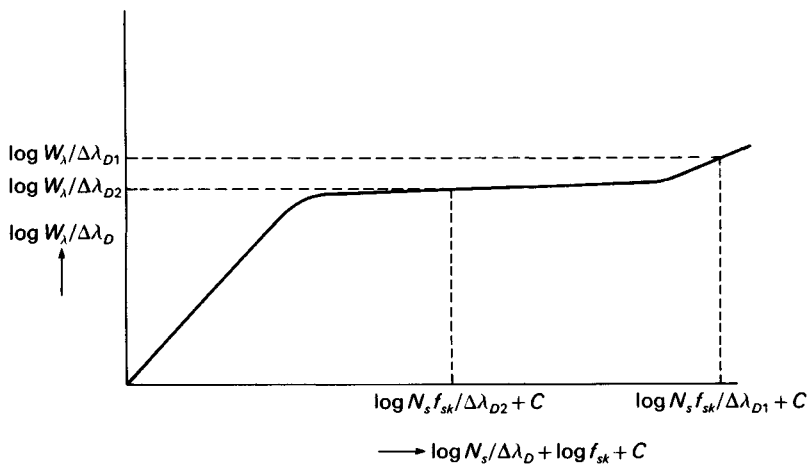

Fig. 12.7 shows a schematic curve of growth. Different values of $\Delta\lambda_D$ may lead to very different values for the derived number N_s of absorbing atoms.

good oscillator strengths is not an easy task and we are still lacking good values of oscillator strengths for many lines of complicated atoms. But once we have reasonably good values for the oscillator strengths, we can fit the lines of a given multiplet on the theoretical curve of growth shown in Fig. 10.16. The abscissa of this curve of growth is the line absorption coefficient per cm^3 in the line center times a factor $\frac{2}{3} \times 1/\kappa_c \times \mathrm{d} \ln B_\lambda/\mathrm{d}\tau_c$ and then the normalizing factors $1/\Delta\lambda_D$ and $1/R_c$, where R_c is the largest possible central depth of a line. If we consider only a narrow spectral region, then all the last factors mentioned are nearly the same for the different lines. For a small wavelength band we only have to consider the different values for the line absorption coefficient when comparing different lines. The line absorption coefficient in the line center is given by a constant κ_0 and the number of absorbing atoms times the oscillator strength, i.e.,

$$\kappa_L = \kappa_0 N(n = n_i) f_{ik}, \qquad 12.9$$

where n_i stands for the quantum numbers of the lower level involved in the transition, and f_{ik} is the oscillator strength for the transition from the level with quantum numbers i to the level with quantum numbers k. If we plot the abscissa in a logarithmic scale, then the difference in the abscissa for different lines is given by

$$\log N(n = n_s) - \log N(n = n_l) + \log f_{sm} - \log f_{lj}$$

if the line transitions are from s to m and from l to j. The number of atoms in the various states of excitation can be described by the Boltzmann formula, namely

$$\frac{N(n = n_s)}{N_0} = \frac{g_s}{g_0} \mathrm{e}^{-\chi_s/kT}, \qquad 12.10$$

where g_s is the statistical weight for the level with quantum numbers s, χ_s is the excitation energy for this level, N_0 is the number of atoms in the ground level, and g_0 is the statistical weight of the ground level. For lines of a given multiplet the χ_s are all the same for different lines because the fine structure splitting of the levels is very small. With this in mind we can then say that the difference in the abscissa for lines of a given multiplet is given by

$$\Delta \text{ abscissa} = \log g_s + \log f_{sm} - \log g_l - \log f_{lj}, \qquad 12.11$$

or

$$\Delta \text{ abscissa} = \log g_s f_{sm} - \log g_l f_{lj}, \qquad 12.12$$

because N_0 and g_0 are also the same for one multiplet. In fact, they are the same for all lines of a given atom or ion.

It is always the product gf which we need and it is also this product which is measured from the line intensities in the laboratory. Therefore, this product is usually given in the oscillator strength tables.

For the lines in a given multiplet we then know the difference in the abscissa. The equivalent widths have to be measured on the spectrum. To begin with, we can plot the first line sm (going from quantum state s to the state m) of the multiplet at an arbitrary position of the abscissa in the plot with the curve of growth, because we do not know the exact value for the abscissa. When we now plot the value for the second line lj we know that its abscissa must be decreased by the amount $\log g_s f_{sm} - \log g_l f_{lj}$, as compared to the abscissa for the first line. We therefore know the relative position of this line with respect to the first line. For the third and fourth lines of the same multiplet we also know the relative values of the abscissa. After we have plotted all the values for the different lines of the multiplet we have a picture, as shown in Fig. 12.8 by the $\times$. Clearly, these points must fit on the theoretical relation between the equivalent widths and the abscissa. We see that we have plotted all our points for the wrong absolute value of the abscissa, which was to be expected, since we chose the abscissa of the first line arbitrarily. Shifting the points over onto the theoretical curve of growth without changing the relative values of the abscissa tells us where these points really belong on the curve of growth.

12.3.3 Determination of $\Delta\lambda_D$

We have still not talked about how to determine the ordinate which is not solely determined by the equivalent width – which we can

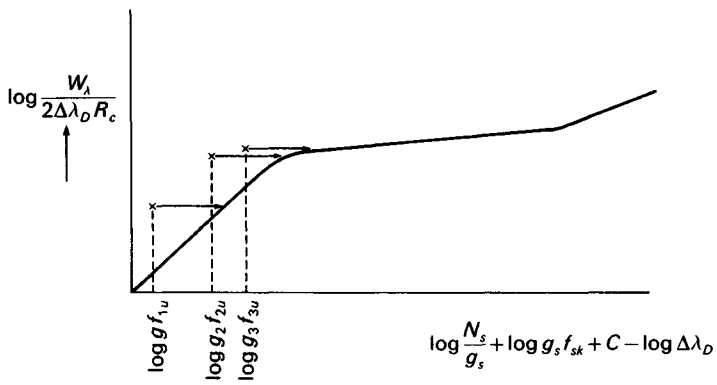

Fig. 12.8. For three lines of one multiplet the values of $\log gf$ are known and therefore their distances on the abscissa are known. The shift over onto the theoretical curve of growth determines the absolute values for the abscissa of the three lines.

measure – but also by the Doppler width $\Delta\lambda_D$ and the largest central depth R_c. On high resolution spectra for slowly rotating stars R_c can be measured directly on the spectrum for the strongest lines in the spectral region being studied. (Remember that R_c is wavelength dependent.) For low resolution spectra and for spectra of rapidly rotating stars this is not possible. We should then take the values measured for similar stars which are slow rotators and for which high resolution spectra are available. The intrinsic R_c are rather similar for different stars even of slightly different spectral types.

For a determination of $\Delta\lambda_D$ we started assuming that we see only thermal Doppler broadening, which means $\Delta\lambda_D = \lambda\xi_{\text{th}}/c$, where c is the velocity of light. If this Doppler broadening is too small, i.e., if there is microturbulence broadening in addition, then the equivalent widths for the lines on the flat part of the curve of growth are larger than corresponding to the thermal $\Delta\lambda_D$. They then appear too large in comparison with the equivalent widths of the weaker lines (see Fig. 12.9). If in a given multiplet we have weak lines *and* strong lines such that the horizontal shift is determined by the weak lines, the error in $\Delta\lambda_D$ can be obtained from the strong lines (as shown in Fig. 12.9). Once the $\Delta\lambda_D$ is determined from one multiplet, the same value can be used for the other multiplets of the same atom or ion in the same spectral region. The absolute value of the abscissa determines the number of atoms $N(n = n_s)$ absorbing in the lines of this multiplet. This number can be read off, except for a factor C, determined by the continuous κ and the gradient of the Planck function. This factor C is, however, the same for all lines in a given small wavelength region.

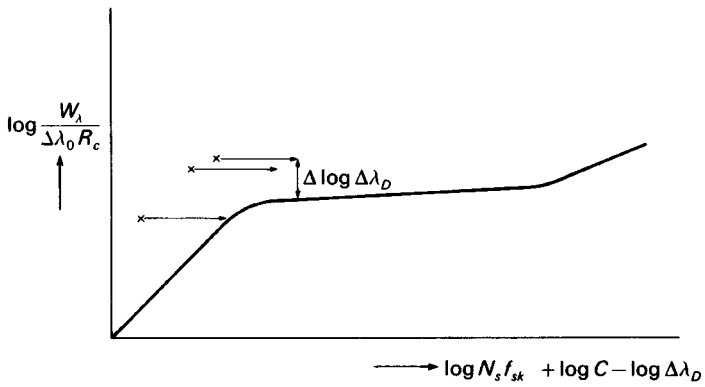

Fig. 12.9. If the assumed value for $\Delta\lambda_D$ was too small the points for the saturated lines lie too high by the error in $\Delta \log \Delta\lambda_D$, when the absolute value for the abscissa is determined by the weak lines.

12.3.4 Determination of the excitation temperature T_{exc}

After we have fitted one multiplet onto the curve of growth, we can proceed to do the same with the next multiplet and determine the number of atoms N_n absorbing in this multiplet with an excitation energy χ_n, except again for the factor C. The ratio of the occupation numbers for the levels s and n can then be determined because the common factor C cancels out. Inserting this ratio into the Boltzmann formula, we find

$$\frac{N_n}{N_s} = \frac{g_n}{g_s} e^{-(\chi_n - \chi_s)/kT_{exc}}. \qquad 12.13$$

The temperature in the exponent is again some average value of the temperature in the atmosphere. This average is taken with different weighting factors than the temperature, which we determined from the Balmer discontinuity and the Balmer lines. Therefore, it does not necessarily have to be exactly the same as the Balmer line temperature, although the differences should never be large; at least not if local thermodynamic equilibrium is a good approximation. In order to make clear how this temperature was determined, we call this temperature the *excitation temperature*, because it is determined from the excitation of the different energy levels. Even if LTE should not be a good approximation, excitation temperatures can always be defined by means of equation (12.13). But the values obtained with occupation numbers for different levels can then be quite different. In fact, the excitation temperatures obtained for different energy levels can serve as one test, whether the assumption of local thermodynamic equilibrium is valid. If the excitation temperatures disagree distinctly, then the assumption of LTE is not a good one.

12.3.5 Determination of the ionization temperature

When we have determined the occupation number N_s of a given level and also the excitation temperature T_{exc}, we are in the position of being able to calculate the occupation numbers for all other levels for that atom or ion for which we cannot observe lines. With this we calculate the total number of atoms of the kind for which we have studied the lines. Again, this number still has the unknown factor C attached. Suppose we have made this study for the neutral iron lines. We can then make the same kind of investigation for the lines of the ionized iron and determine the total number of Fe^+ ions. If we know what the electron pressure is – for example, from the Balmer discontinuity or from the Inglis–Teller formula – we can use the Saha equation, plug in the ratio of Fe^+/Fe and

determine the temperature from the Saha equation. Because this temperature is determined from the degree of ionization, it is called the *ionization temperature*. Again, this is an average over the photosphere with slightly different weights in the averaging procedure. This temperature may be slightly different from the excitation temperature. We can go one step further and do the same procedure for another element, for instance, titanium. For titanium the $\Delta\lambda_D$ is probably different from the value obtained for iron because its atomic weight is different. Therefore, the thermal velocity is different. We have to determine a new $\Delta\lambda_D$ for titanium. So we again find a multiplet with many lines of different equivalent widths such that we can determine a curve of growth from one multiplet. If we do not find a suitable one we can actually combine several multiplets. We know that the occupation numbers of the lower levels are determined by the Boltzmann formula. If the first multiplet originates from a level with quantum number n and excitation energy χ_n, and the other multiplet originates from a level with quantum number m and excitation energy χ_m, then we know from the Boltzmann formula that the ratio of the occupation numbers N_n and N_m is given by

$$\log N_m - \log N_n = \log g_m - \log g_n - \Theta(\chi_m - \chi_n), \qquad 12.14$$

or

$$\log \frac{N_m}{g_m} - \log \frac{N_n}{g_n} = -\Theta(\chi_m - \chi_n). \qquad 12.15$$

If we use the number of absorbing atoms in the level n, namely N_n, in the abscissa of the curve of growth for both multiplets, then we have to express the number of atoms N_m absorbing in the second multiplet by means of N_n and the Boltzmann formula, as given by equation (12.15), which gives the difference in abscissa for the two multiplets. Unfortunately, there is the quantity $\Theta = 5040/T_{\text{exc}}$ in equation (12.15), and we do not yet know its value. We can, however, try different values of Θ and see for which Θ the two multiplets fit best on one curve. If the assumption of LTE is valid, then the same Θ should also hold for other multiplets. We can therefore use as many multiplets as we want and determine Θ from the condition that all the multiplets should fit on one curve of growth with the smallest amount of scatter. This can also be done graphically, as demonstrated in Fig. 12.10. As previously shown, we plot $\log(W_\lambda/(\Delta\lambda_D R_c)$ as a function of $\log(gf)_{ni}$ for the different lines with index i of the multiplet from level n. We then do the same for the lines with index j from the level m. Since $\log N_m/g_m$ and $\log N_n/g_n$ differ by $\Theta(\chi_m - \chi_n)$ as given by equation (12.15), the abscissae for the two multiplets differ by this amount, which

means the two curves of growth outlined by the two multiplets are shifted with respect to each other by this amount. The horizontal shift necessary to fit one multiplet on top of the other determines $\Theta(\chi_m - \chi_n)$. With the known excitation energies the value of Θ can be obtained.

The excitation temperature can thus be obtained by fitting the different multiplets together.

12.3.6 The determination of the kinetic temperature

In Section 12.3.3 we saw that $\Delta\lambda_D$ – the Doppler width – given by the thermal motions and by the microturbulent motions, can be determined from the vertical shift necessary to fit the observed curve of growth onto the theoretical curve of growth. We also emphasized that the value for $\Delta\lambda_D$ must be expected to be different for different elements with different atomic weights, because their thermal velocities are different. In principle, this difference in thermal motion for different particles can be used to determine the kinetic temperature in the gas, which we would generally consider to be *the* temperature of the gas.

We saw in Section 10.1.5 that $\Delta\lambda_D$ is given by

$$\Delta\lambda_D/\lambda = \xi/c \qquad \text{with } \xi^2 = \xi_{\text{th}}^2 + \xi_{\text{turb}}^2, \tag{12.16}$$

where ξ_{turb} is the microturbulent reference velocity and where the thermal reference velocity ξ_{th} is given by

$$\xi_{\text{th}}^2 = \frac{2R_g T}{\mu}, \tag{12.17}$$

μ = atomic weight.

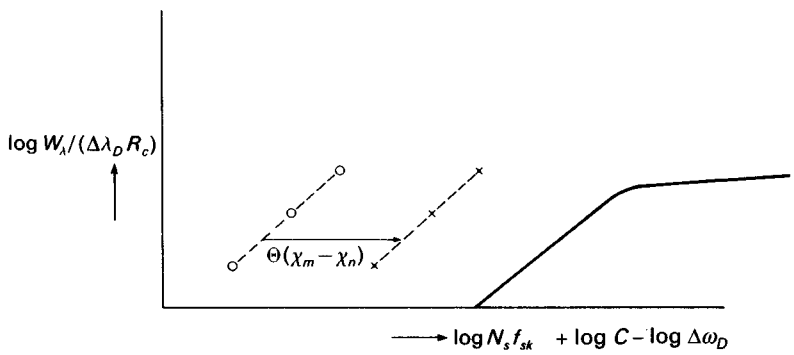

Fig. 12.10. The equivalent width for the lines in multiplet n and m divided by $\Delta\lambda_D R_c$ are plotted as a function of log $g_{ni}f_{ni}$ and log $g_{mj}f_{mj}$, respectively. Their abscissae in the curve of growth differ by $\log N_m/g_m - \log N_n/g_n = -\Theta(\chi_m - \chi_n)$ which can be determined by the horizontal shift necessary to shift the points for the two multiplets on top of each other.

For particles with large μ the thermal velocity is small, while particles with a small molecular weight μ have large velocities. It is this difference which, in principle, makes it possible to determine the kinetic temperature. Suppose we have measured $\Delta\lambda_D$ for iron lines and for carbon lines. We then know $\xi^2(\mathrm{Fe})$ and $\xi^2(\mathrm{C})$, where

$$\xi^2(\mathrm{Fe}) = \frac{2R_g T}{\mu(\mathrm{Fe})} + \xi^2_{\mathrm{turb}}, \qquad 12.18$$

and

$$\xi^2(\mathrm{C}) = \frac{2R_g T}{\mu(\mathrm{C})} + \xi^2_{\mathrm{turb}}. \qquad 12.19$$

From equations (12.18) and (12.19) we then find

$$T = \left(\frac{\xi^2(\mathrm{Fe})}{2R_g} - \frac{\xi^2(\mathrm{C})}{2R_g}\right)\left(\frac{1}{\mu(\mathrm{Fe})} - \frac{1}{\mu(\mathrm{C})}\right)^{-1}. \qquad 12.20$$

In practice, there are limits to the accuracy of this temperature determination. First of all, this can only work if the microturbulent velocities ξ_{turb} are not much larger than the thermal velocities. Otherwise, the differences in $\Delta\lambda_D$ are too small to be measurable. The second problem is that the atomic data, as well as the measured equivalent widths, are often not very accurately known, which means that the best vertical shift between observed and theoretical curve of growth is often not very accurately determined, resulting in rather large uncertainties for $\Delta\lambda_D$ and therefore for ξ^2. However, in the future improved atomic data and higher accuracy of the measurements will make this determination of a kinetic temperature a nice tool to check on our assumption of LTE.

12.3.7 The determination of the electron pressure from the curve of growth

In Section 12.3.5 we discussed how we can determine the ionization temperature of, for instance, iron, if from the curve of growth analysis we have determined both the number of neutral iron atoms Fe and the number of ionized iron ions Fe^+. We stated that, having obtained the number of free electrons n_e from a study of the Balmer lines and/or the Balmer discontinuity, we can use the Saha equation to calculate the ionization temperature T_{ion}. Here we show that the electron density n_e can, in principle, also be determined from the curve of growth analysis. Suppose we have not only determined the number of iron atoms and ions, but also the number of titanium atoms and ions. We can then use the Saha equations for both elements, which gives us two equations to determine both

unknowns: the ionization temperature and the electron density. The division of both Saha equations gives one equation for the determination of the ionization temperature:

$$\log(\mathrm{Fe}^+/\mathrm{Fe}) - \log(\mathrm{Ti}^+/\mathrm{Ti}) = \log(u^+/u)_{\mathrm{Fe}} - \log(u^+/u)_{\mathrm{Ti}} - \Theta(\chi_{\mathrm{ion}}(\mathrm{Fe}) - \chi_{\mathrm{ion}}(\mathrm{Ti})). \quad 12.21$$

Once the ionization temperature, or Θ, has been determined from this equation, the electron density can be obtained from either of the two Saha equations.

As is obvious from equation (12.21), the determination of Θ, and therefore the ionization temperature, by this method is very inaccurate if the ionization energies of the elements being used are nearly equal. Small errors in the empirical determination of the left-hand side of the equation lead then to very large errors in the value for Θ and T_{ion}. When using this method we should therefore always try to use element pairs with very different ionization energies. Unfortunately, this is not always possible, because for elements with very low ionization energies the atomic lines may no longer be visible, while for those with high ionization energies the ionic lines are too weak to be measurable. The elements or ions observable in both stages of ionization usually have ionization energies which do not differ very much. Therefore, it is often preferable to determine the electron density from the Balmer lines and discontinuity and determine the ionization temperature from a single Saha equation.

12.3.8 Dependence of some line strengths on temperature, gravity and element abundances

In Volume 1 we pointed out that the temperature of a star has a large influence on the strengths of the different spectral lines. For high temperature stars there are no lines of neutral elements in the spectrum because the atoms are all ionized except for helium, which has a very high ionization energy, and except for hydrogen, which is so very abundant and which also determines the continuous absorption coefficient. The line depth R_λ depends on the ratio of the line absorption coefficient to the continuous absorption coefficient, which for the Balmer lines means

$$R_\lambda \propto \frac{\kappa_L}{\kappa_c} \propto \frac{N_{\mathrm{H}}(n=2)}{N_{\mathrm{H}}(n=3)} n_e = \frac{g_2}{g_3} \mathrm{e}^{-(\chi_2-\chi_3)/kT} n_e = \frac{g_2}{g_3} \mathrm{e}^{(\chi_3-\chi_2)/kT} n_e, \quad 12.22$$

if the continuous absorption is due to the neutral hydrogen atoms in the third quantum level, as, for example, in early B stars. For increasing T the R_λ decreases. For increasing degree of ionization of hydrogen, both the line absorption and the continuous absorption coefficients decrease and the ratio

only changes because the occupation number in the third quantum level, doing the continuous absorption, increases relative to the one of the second quantum level responsible for the line absorption. This is the reason for the decrease of the Balmer line intensities with increasing temperature for the B stars. The temperature dependence is very weak because $\chi_3 - \chi_2$ is so small.

When comparing line strengths in supergiants with those of dwarfs, we have to be aware of the influence of the gravity which determines the pressure in the atmospheres. For instance, for G type stars the continuous absorption is due to the H^- ion. The Saha equation tells us that the number of H^- ions relative to the number of neutral atoms increases with increasing electron density n_e. Supergiants therefore generally show stronger spectral lines than dwarf stars (see Fig. 12.1), not because there are more heavy elements in their atmospheres, but because the continuous absorption coefficient is smaller in these atmospheres of low gravity and therefore small electron density. Therefore, we have to be very cautious when looking at stellar spectra. Weak lines do not necessarily mean a low abundance of heavy elements and vice versa.

For the Balmer lines of hydrogen we find, for instance, for A and B stars that the lines become *stronger* when the *hydrogen abundance decreases* if the hydrogen is replaced by helium. The reason is the increase in gas pressure coupled with an increase in the electron pressure in the atmospheres with larger helium abundances. Because the line absorption coefficient in the Balmer line wings is proportional to the electron density and the continuous absorption coefficient is still due to the hydrogen, we find an increase of the hydrogen line strength for stars with a smaller hydrogen abundance, unless we go to extremely small hydrogen abundances, when the continuous κ becomes due to helium.

When making a computer analysis of stellar spectra, we have to be aware that when trying different abundances of heavy elements in order to match the observed spectrum we have to be sure to change the atmospheric model each time we try a different element abundance because the change in the continuous absorption coefficient due to the changing metal abundance is as important as the change in the line absorption coefficient.

As an example, let us look at the dependence of an Fe^+ line strength on the abundance of Fe for solar type stars. For such a line the line depth is determined by the ratio

$$R_\lambda(\mathrm{Fe}^+) \propto \frac{\mathrm{Fe}^+}{\mathrm{H}^-} \propto \frac{\mathrm{Fe}^+}{N_\mathrm{H} n_e}. \qquad 12.23$$

In solar type stars the electrons come from the ionization of the metals like iron. We saw that in the sun iron is mainly ionized. The iron abundance

is therefore given by

$$\text{abundance (Fe)} = \text{Fe}^+/N_\text{H} = A(\text{Fe}). \qquad 12.24$$

If the abundances of all the heavy elements vary together, as is roughly the case, then we must also have

$$n_e \propto \text{Fe}^+ = A(\text{Fe})N_\text{H}. \qquad 12.25$$

Inserting this into equation (12.23), we find

$$R_\lambda(\text{Fe}^+) \propto \frac{A(\text{Fe})N_\text{H}}{N_\text{H}n_e} = \frac{A(\text{Fe})}{n_e} \propto \frac{1}{N_\text{H}}. \qquad 12.26$$

The Fe abundance cancels out in a first approximation. Actually, N_H increases for decreasing metal abundances. Therefore, we do find a slight decrease in the line strengths of the Fe^+ lines for decreasing metal abundances, but the effect is very small. A rough glance at the spectrum may be very misleading.

12.4 Observed element abundances

12.4.1 Population I stars

With the curve of growth analysis as described in the previous chapter, we can determine abundances for all elements for which spectral lines can be identified and for which the necessary atomic data have been determined, either by laboratory experiments or by theoretical studies. We can study all stars which are bright enough to be observed with a dispersion high enough such that the different lines can be separated at least somewhere in the spectrum. When comparing stars of very different temperatures, the comparison of the abundances in the stars is usually accurate to within a factor of 2. When comparing stars of nearly equal temperatures and gravities, accuracies of about 25% can be reached in a very careful relative analysis. When higher accuracies are claimed this is usually misleading.

The stellar analyses show that almost all of the nearby stars have the same relative element abundances to within the uncertainty factor of about a factor of two. It is found that the vast majority of stars consists mainly of hydrogen with about 10% ± 5% of helium (by number of atoms and ions). Since each helium nucleus is four times as heavy as a proton, the abundance of helium by weight, called Y, can be computed from

$$Y = \frac{4 \times 0.1}{0.9 + 0.4} = 0.31\ (\pm 0.1). \qquad 12.27$$

Helium lines can only be seen in the spectra of hot stars. Therefore, the helium abundance can only be determined for these stars. In the cool stars

Table 12.3. *Abundances* log N *of the most abundant elements*[a]

		Population I		Population II		
Atomic number	Element	Sun[b] G2 V	α Lyr A0 V	γ Ser F6 IV	HD 140283	HD 19445
1	Hydrogen (H)	12.0	12.0	12.0	12.0	12.0
2	Helium (He)	(11.0)	(11.4)	?	?	?
8	Oxygen (O)	8.8	9.3	9.1	?	?
6	Carbon (C)	8.5	?	8.4	6.4	?
7	Nitrogen (N)	8.0	8.8	?	?	?
10	Neon (Ne)	(7.9)	?	?	?	?
26	Iron (Fe)	7.6	7.1	7.2	5.2 ± 0.3	5.7 ± 0.2
14	Silicon (Si)	7.5	8.2	7.4	5.1	6.0
12	Magnesium (Mg)	7.4	7.7	7.5	4.9	6.5?
16	Sulfur (S)	7.2	?	7.2	?	?
13	Aluminum (Al)	6.4	5.7	6.1	3.7	4.5
28	Nickel (Ni)	6.3	7.0	6.4	4.2	4.7
20	Calcium (Ca)	6.3	6.3	5.9	4.0	4.8
11	Sodium (Na)	6.3	7.3	6.1	3.5	?
24	Chromium (Cr)	5.9	5.6	4.9	3.6	3.8
17	Chlorine (Cl)	5.6	?	?	?	?
15	Phosphorus (P)	5.5	?	?	?	?
25	Manganese (Mn)	5.4	5.3	4.7	2.8	3.7
22	Titanium (Ti)	5.1	4.7	4.3	2.8	3.4
27	Cobalt (Co)	5.1	?	4.1	2.7	?
19	Potassium (K)	5.0	?	?	?	?

[a]Normalized to log $N(\mathrm{H}) = 12.00$.
[b]These abundances are considered to be 'cosmic abundances'.
Values in parentheses are uncertain.

we can see helium lines only in the chromospheric spectrum for which analysis is much more difficult and uncertain because LTE cannot be assumed. We have very incomplete knowledge about the helium abundance in cool stars and must resort to indirect methods which will be discussed in Volume 3. Unfortunately, the accurate knowledge of the helium abundance in old stars, which are all cool stars, is very important for many cosmological problems.

There is only a minor trace of heavy elements mixed in with the hydrogen and helium – only about 10^{-3} by number of atoms – but because these elements are heavy their fraction by mass, Z, still amounts to about 2%, i.e., $Z \approx 0.02$. This leaves for the hydrogen abundance by weight X

$$X = 0.67 \pm 0.1$$

for most of the 'normal' stars.

Among the heavy elements the most abundant ones are C, N, O, Ne. For each of these atoms or ions there are about 1000 hydrogen atoms or ions. The elements next in abundances are Fe, Si, Al, and Mg, but they are less abundant again by another factor of 10. Table 12.3 gives recent abundance determinations for the more abundant elements for stars in our neighborhood, the so-called Population I stars.

12.4.2 Element abundances in Population II stars

There are a few stars in our neighborhood, however, which show distinctly different abundances. They were originally noticed as subdwarfs (in the color magnitude diagram, they appear below the main sequence) or as weak lined stars. The stars generally also show large velocities relative to the sun and were therefore also called high velocity stars. We call these generally the Population II stars. These stars still mainly consist of hydrogen and helium, probably in almost the same proportions as in the Population I stars, but the heavy element abundances are generally lower by factors ranging from 3 to about 200. Surprisingly, the relative abundances within the group of heavy elements varies very little between Population I and Population II stars, although there is recent evidence that for intermediate degrees of depletion of the iron group elements the lighter elements are less depleted than the iron group elements. Also, variations in the ratio of the C, N, and O abundances are observed for various groups of stars. These details will be discussed in Volume 3, where we will study the origin of the chemical elements.

Element abundances for some high velocity Population II stars are also given in Table 12.3.

The depletion of heavy elements increases generally with increasing distance of the stars from the galactic plane. Those few high velocity stars in our neighborhood show by their velocities that they are actually on their way out to large distances in the halo again, from where they originally came. They just happen to pass by in their galactic orbit.